

Order Recovery and the Reorderability Law

A finite index is closed only relative to an order on each of its coordinates; when those orders are unknown it is decidable whether any of them closes the index, the orders are then recoverable — exactly at two coordinates by the totality of a “may precede” relation, and in general by propagation along the constraint tree — and with every order unknown the constraint a pair of cells imposes on the orientation variables is bijunctive exactly when it is a partition constraint, which places the first non-bijunctive constraint at three coordinates.

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Abstract

A finite index is a set of tuples over finite alphabets. It is *closed* when it contains the coordinatewise meet and join of any two of its cells — but meet and join are defined only once each alphabet has been given a linear order, so closure is a property of the pair (index, ordering) and never of the index alone. This paper asks what an index can say about its own ordering. Three results answer it. First, closure is equivalent to the vanishing of the staircase defect E : an index is closed exactly when it is the fixed point of its own monotone envelopes, which gives an index its alphabet, its bounds and its membership test from its cells alone. Second, define $u \sqsubseteq v$ on one coordinate when every meet of a cell over u with a cell over v lies over u and every join lies over v . The relation $\sqsubseteq$ is transitive for every index whatever, and an order s on that coordinate closes the index exactly when $\sqsubseteq$ is reflexive and contains s ; so the index is closable on that coordinate exactly when $\sqsubseteq$ is a total preorder, and the admissible orders are precisely the linear extensions of $\sqsubseteq$. At two coordinates this is a complete criterion, equivalent to the statement that every fibre is an interval and no fibre nests strictly inside another. Above two coordinates the pointwise criterion is necessary and not sufficient, but when the constraint graph is a tree the joint condition factorises over its edges, and a memoised dynamic program rooted anywhere recovers every admissible ordering system with at most $\sum_i |A_i|!$ evaluations — the sum of the alphabets’ permutation counts rather than their product. Third, with all orders unknown the problem is a constraint system on orientation variables; the constraint a pair of cells imposes is invariant under reversing all of its own orientations and contains the two constant assignments, and every such relation arises. Such a constraint is bijunctive — closed under the ternary majority, hence expressible by two-clauses — exactly when it is a partition constraint, “the orientations agree on each block of a partition of the coordinates on which the cells differ”; the bijunctive constraints at arity k therefore number the Bell number $B(k)$: 2, 5, 15, 52 of 2, 8, 128, 32,768. The first non-bijunctive constraint appears at three coordinates, with an explicit six-cell witness, and counting bijunctivity on the auxiliary difference relation instead — where every relation at arity three is majority-closed — has no two-clause consequence, because an exclusive-or is not bijunctive. The growth step of the reorderable family is 2^{d-2} at every censused box of dimension $d \leq 4$, and at the full Boolean box the drop is 2^{d-2} for every d , from a tight bound on proper sublattices: $3 \cdot 2^{d-2}$, three quarters, which every maximal sublattice attains. Every finite claim is recomputed: 111 obligations, 28 of them machine-checked by Z3 over every subset of a named box that uses every value and, where an order is unknown, over every total order on the named coordinate. The electron-configuration index Λ is the worked case throughout; its eight orders are recovered from a bag of 976 cells with all eight alphabets independently permuted, and the recovery is exact up to the 16 orderings that admit Λ at all, out of 11,943,936.

§0 · The result

A closed index states its own alphabet and its own bounds unconditionally, and its own order under a stated condition on the constraint graph.

An index is a finite set X of tuples. Write A_i for the set of values its i -th coordinate takes. If each A_i carries a linear order, X sits inside a product of chains, and X is **closed** when it is a sublattice of that product — closed under coordinatewise minimum and maximum. Nothing in the cells of X fixes those orders. The orders are a choice made when the index is drawn, and the question of this paper is how much of that choice the cells themselves determine.

What is established.

1. **Closure is self-expressing** (Theorem 1, PROVED; EXHAUSTIVE on 70,598 subsets). For each ordered pair of coordinates let $\phi_{ij}(v) = \max \{ x_i : x \in X, x_j \leq v \}$ — the tightest monotone bound the cells support — and let $\mathcal{R}(X)$ be the set of ambient points respecting all $d(d-1)$ of them, $E(X) = |\mathcal{R}(X)| - |X|$. Then $E(X) = 0$ if and only if X is closed. A closed index is therefore *equal to* the region its own envelopes cut out: its membership test is a finite conjunction of comparisons read off its cells, and is total on the ambient box.
2. **The order relation, and what it decides** (Theorem 2, PROVED and MACHINE-CHECKED over boxes 3×3 , 4×4 , $2 \times 2 \times 3$ and $3 \times 3 \times 3$). Fix orders on every coordinate but one. Say $u \sqsubseteq v$ when, for every cell over u and every cell over v , the meet of their remaining coordinates lies over u and the join lies over v . Then (a) $\sqsubseteq$ is transitive for every index, with no hypothesis at all; (b) an order s on that coordinate closes X exactly when $\sqsubseteq$ is reflexive and $u <_s v \Rightarrow u \sqsubseteq v$; (c) some order closes X exactly when $\sqsubseteq$ is a total preorder, and then every linear extension of $\sqsubseteq$ is such an order and nothing else is. The admissible orders on one coordinate are thus *exactly* the linear extensions of a total preorder computable with $|X|^2$ meets and joins, $O(d \cdot |X|^2)$ operations — the recovered order is unique up to the ties of $\sqsubseteq$, and the ties are the coordinate values the index cannot tell apart.
3. **Two coordinates: a complete criterion** (Theorem 3, PROVED and MACHINE-CHECKED over boxes 3×3 , 4×4 , 3×5 and 5×4 ; Lemma 5 EXHAUSTIVE over 2,354 families of distinct intervals and 5,084 multisets). At $d = 2$, X is closed under s if and only if every fibre is an interval of the other chain and both endpoints are non-decreasing along s . A family of intervals admits such an order if and only if no interval nests **strictly** inside another; sorting by left endpoint, ties by right, decides it. *Strictly* is load-bearing: reading it as ordinary containment refutes the lemma on 1,098 of the 2,354 families.
4. **Above two coordinates, propagation** (Theorems 4 and 5, PROVED; EXHAUSTIVE on 7,486 tree-structured subsets and 362,144 scrambles). Every pair projection being reorderable is necessary and not sufficient: the three unit vectors of $\{0,1\}^3$ refute the converse, and the refutation is proved, not merely found (§4.1). But when X is the intersection of the cylinders over the projections onto the edges of a graph, closure factorises: X is closed under an ordering system if and only if every edge projection is. When the graph is a tree, a dynamic program rooted at any coordinate returns **every** admissible ordering system, trying $|A_r|!$ orders at the root and solving each further coordinate from its parent by Theorem 2; memoised on the pair (coordinate, order), it makes at most $\sum_i |A_i|!$ evaluations. On Λ that is 94 against $\prod_i |A_i|! = 11,943,936$.
5. **The reorderability law** (Theorem 8, PROVED; EXHAUSTIVE on 1,967,872 pair constraints and on every pair constraint of four arities). With every order unknown, the decision is a constraint system on orientation variables. The constraint a pair of cells imposes is invariant under reversing all of its own orientations and contains the two constant assignments, and every such relation arises (Theorem 7, PROVED). Such a constraint is bijunctive — closed under the ternary majority, hence a conjunction of two-clauses — if and only if it is a **partition constraint**: the orientations agree on each block of a partition of the coordinates on which the two cells differ. The bijunctive constraints at arity k are therefore in bijection with the partitions of a k -set and number the Bell number $B(k)$: 2 of 2 at $k = 2$, 5 of 8 at $k = 3$, 15 of 128 at $k = 4$, 52

of 32,768 at $k = 5$. Since $k \leq d$ and $k = d$ is attained, **an index of two coordinates imposes only bijunctive constraints on its orientation variables, and the first non-bijunctive constraint appears at $d = 3$** : the six cells $\{000, 111, 001, 110, 010, 101\}$ impose, at the pair $(000, 111)$, the constraint “ $\sigma_1 = \sigma_2$ or $\sigma_1 = \sigma_3$ ”, and the majority of $001, 010, 111$ is 011 , which it excludes. Counted on the auxiliary difference relation of Lemma 7 the figures are 2 of 2, 8 of 8, 73 of 128 and 1,442 of 32,768 — every difference relation at arity ≤ 3 is majority-closed — but bijunctivity of the difference relation has no two-clause consequence for the orientation variables, because a difference is an exclusive-or of two of them and exclusive-or is not bijunctive.

6. **The growth step** (Theorem 9, PROVED at the Boolean box, EXHAUSTIVE over ten boxes). Every proper sublattice of the Boolean box 2^d has at most $3 \cdot 2^{d-2}$ cells, three quarters, and that is attained by the cells satisfying one implication between two coordinates; the removed cells are a subcube; every maximal sublattice is of that form (Corollary 2, from Rival 1973). Hence the drop at the full box — the least number of cells whose removal leaves a reorderable set — is exactly 2^{d-2} , and the census over ten boxes finds the growth step equal to it everywhere: 1 at $d = 2$, 2 at $d = 3$, 4 at $d = 4$, attained at the full box in every one.

What is not established.

- **No hardness result.** Every pair constraint contains both constant assignments, so the language of pair constraints on its own is 0-valid and 1-valid — two of Schaefer’s tractable classes — and trivially satisfiable. It is antisymmetry, $\alpha_i(u,v) \neq \alpha_i(v,u)$, that is neither; together with the arity-3 witness, which is closed under neither majority, conjunction, disjunction nor ternary exclusive-or, the language lies in none of Schaefer’s six classes, so *satisfiability for that language* is NP-complete. That is not a hardness result for reorderability: it would require a reduction realising arbitrary instances of the language as pairs of cells of an index, and no such reduction is given here. The gap is named, situated against Green and Cohen (2008) and Jeavons and Cooper (1995), and left open (§6.5).
- **The step law above $d = 4$.** The census is exhaustive at $d \leq 4$. At $d = 5$ and $d = 6$ only the drop at the full box is established, from the sublattice bound, and it is a lower bound on the step; the maximum over all reorderable sets at those dimensions is not computed.
- **The general bound on proper sublattices** is proved here for the Boolean box only. For an arbitrary finite distributive lattice, Rival (1973), CITED, characterises the complement of a *maximal* sublattice as an interval and bounds the maximal sublattice from below, $|L| \leq (3/2)|K|$; Theorem 9 bounds every proper sublattice of the Boolean box from above, and the two are not comparable. In the Boolean box every maximal sublattice is exactly three quarters (Corollary 2).
- **The diagnostic is a decomposition, not a repair.** Theorem 6 says that a defect with no pairwise witness must have a ternary one, and names it. It does not say which re-coordination removes a defect, only that reorderability decides the one operation of relabelling.
- **Λ ’s recovery is exact up to the orderings that admit Λ at all.** There are 16 of them at each cap setting, and they are different things at the two: at 976 cells, one global reversal times three ties between values no cell distinguishes; at 216 cells, one tie times the independent reversal of three components into which the constraint tree falls (§5.3). Over 20 seeded scrambles at each setting the recovery returned an admissible ordering 20 times out of 20 — but the *true* ordering or its reverse only 3 times out of 20 at the larger setting and 0 of 20 at the smaller. An index determines its order only up to the orderings under which it is closed, and that is the correct statement of what is recovered.

Status words. Six are used and never merged.

status	meaning
PROVED	a proof is written out below and every step is justified

status	meaning
MACHINE-CHECKED	Z3 returned unsat on the negation of an obligation whose variables range over every subset of a named finite box that uses every value of every alphabet (the convention of D1) and, where an order is unknown, over every total order on the named coordinate; both guards passed
EXHAUSTIVE	a decision procedure visited every case in a stated finite family, and the family is named with its size
SAMPLED	a seeded pseudorandom sweep of a stated size; never exhaustive
CITED	taken from the literature, with the full source
REFUTATION	a claim disproved by an explicit witness, and the witness is printed

A machine-checked claim names its box. A sampled figure names its size and seed. Where a result is proved for a general d and checked for small d , both are stated and the reader is told which is which.

§1 • Definitions

D1 (index, alphabet, box). Let $d \geq 2$. An **index** is a finite non-empty set $X \subseteq \prod_i A_i$ of **cells**, where A_i is a finite set of **values**, the i -th **alphabet**. Throughout, and without further mention, $A_i = \{ x_i : x \in X \}$ — every value of every alphabet occurs in some cell. The **box** of X is the product $\prod_i |A_i|$, and an **ambient point** is an element of $\prod_i A_i$. A cell is an ambient point; the converse is what is at issue.

The convention $A_i = \{ x_i : x \in X \}$ is not cosmetic. Two of the results below — the interval property of Theorem 3 and the transitivity of $\sqsubseteq$ in Theorem 2 — are false without it, and §2.4 and §3.1 exhibit why.

D2 (an ordering of a coordinate; an ordering system). An **ordering** of coordinate i is a linear order s_i on A_i , equivalently a bijection $\text{rank}_i : A_i \rightarrow \{0, \dots, |A_i| - 1\}$. An **ordering system** is a tuple $s = (s_i : i = 1, \dots, d)$, one ordering per coordinate. An ordering system makes each A_i a chain and the ambient box the product of those chains, which is a finite distributive lattice (Davey and Priestley 2002, CITED) under

$$(x \wedge_s y)_i = \text{the } s_i\text{-smaller of } x_i \text{ and } y_i, (x \vee_s y)_i = \text{the } s_i\text{-larger.}$$

The number of ordering systems is $\prod_i |A_i|!$. The **reverse** of s is the system reversing every s_i ; it exchanges $\wedge_s$ with $\vee_s$.

D3 (closure relative to an ordering). X is **closed under** s if $x \wedge_s y \in X$ and $x \vee_s y \in X$ for all $x, y \in X$ — that is, if X is a sublattice of the product of chains that s defines. X is **reorderable** if it is closed under some ordering system. Closure is a property of the pair (X, s) . Since reversing every coordinate exchanges the two operations, X is closed under s if and only if it is closed under the reverse of s ; admissible ordering systems come in reverse pairs.

D4 (envelope, recovery operator, defect). For $i \neq j$ and $v \in A_j$, the **envelope** is

$$\varphi_{ij}(v) = \max \{ x_i : x \in X, x_j \leq v \},$$

the maximum and the comparison taken in the given ordering; under D1 the set is never empty, since v is the j -th coordinate of some cell. The **recovery operator** is

$$\mathcal{R}(X) = \{ x \in \prod_i A_i : x_i \leq \varphi_{ij}(x_j) \text{ for all } i \neq j \},$$

and the **defect** is $E(X) = |\mathcal{R}(X)| - |X|$. That $\mathcal{R}$ is extensive, monotone and idempotent — a closure operator in the sense of Moore (1910) and Ward (1942), CITED — and that its constraints are the monotone staircases of Deville, Barette and Van Hentenryck (1999) and the row-convex constraints of van Beek and Dechter (1995), CITED, are stated and proved in the companion paper on the closure law and are not re-derived here. What this paper needs of $\mathcal{R}$ is $X \subseteq \mathcal{R}(X)$, which is immediate ($x \in X$ gives $x_i \leq \phi_{ij}(x_j)$ by taking the witness x itself), and Theorem 1.

D5 (fibre). For a coordinate i and a value $u \in A_i$, the **fibre** over u is

$$F_i(u) = \{ x_{-i} : x \in X, x_i = u \} \subseteq \prod_{j \neq i} A_j,$$

where x_{-i} deletes the i -th entry. Every fibre is non-empty, by D1. When the coordinates other than i are ordered, $\prod_{j \neq i} A_j$ is a lattice and meets and joins of fibre elements are taken there. The subscript is dropped when the coordinate is clear. At $d = 2$ the fibre over $u \in A_0$ is a subset of the chain A_1 and is written $F(u)$; it is an **interval** if $b_1, b_3 \in F(u)$ and $b_1 \leq b_2 \leq b_3$ imply $b_2 \in F(u)$, and then $lo(u)$ and $hi(u)$ denote its endpoints.

D6 (may precede). Fix a coordinate i and orderings of all the others. For $u, v \in A_i$, say **u may precede v** , written $u \sqsubseteq v$, when

$$\text{for all } p \in F(u) \text{ and all } q \in F(v): p \wedge q \in F(u) \text{ and } p \vee q \in F(v).$$

$\sqsubseteq$ depends on X and on the orderings of the other coordinates, and on nothing else; in particular it does not presuppose an ordering of A_i . It is **reflexive** if $u \sqsubseteq u$ for every u — equivalently, if every fibre is a sublattice — and **total** if $u \sqsubseteq v$ or $v \sqsubseteq u$ for every pair. A reflexive, total, transitive relation is a **total preorder**; its **strict part** is $u < v$ when $u \sqsubseteq v$ and not $v \sqsubseteq u$, and a **linear extension** of it is a linear order s with $u <_s v \Rightarrow u \sqsubseteq v$.

D7 (the recovered order). Given a total preorder $\sqsubseteq$ on A_i , the **recovered order** s^* sorts A_i by the number of strict $\sqsubseteq$ -predecessors, $\text{rank}(u) = |\{ v : v < u \}|$, ties broken by any fixed rule on the raw values. Lemma 4 shows s^* is a linear extension of $\sqsubseteq$ and that the linear extensions are exactly the admissible orders.

D8 (constraint graph; G-structured, tree-structured). For $i < j$, $\pi_{ij}(X) = \{ (x_i, x_j) : x \in X \}$. A pair $\{i, j\}$ is **binding** if $|\pi_{ij}(X)| < |A_i| \cdot |A_j|$. Let G be a graph on the vertex set $\{1, \dots, d\}$. X is **G-structured** if

$$X = \{ x \in \prod_i A_i : (x_i, x_j) \in \pi_{ij}(X) \text{ for every edge } (i, j) \text{ of } G \},$$

that is, if X is exactly the intersection of the cylinders over its G -edge projections — equivalently, if X is the solution set of a binary constraint network with graph G whose constraints are its own edge projections. When G is a tree T , X is **T-structured**. Freuder (1982), CITED, proves that a binary network whose graph is a tree is solved backtrack-free once it is arc consistent; that is the property the propagation of §4.3 exploits, and T-structure is the hypothesis under which the network's solution set is X itself.

D9 (orientation variables, and the constraint of a pair). Fix X with every ordering unknown. For a coordinate i and an ordered pair $u \neq v$ in A_i , let $o_i(u, v)$ be a Boolean variable, true when u precedes v . An ordering system is a truth assignment satisfying, on each coordinate, antisymmetry $o_i(u, v) = \neg o_i(v, u)$ and transitivity. For $x, y \in X$ let $I(x, y) = \{ i : x_i \neq y_i \}$ and $k = |I(x, y)|$, the **arity** of the pair. Given an assignment, write $\sigma \in \{0, 1\}^I$ with $\sigma_i = 1$ when x_i precedes y_i . The meet and join of x and y are then determined by σ , and the **constraint of the pair** is

$$C(x, y) = \{ \sigma \in \{0, 1\}^I : \text{the meet and the join determined by } \sigma \text{ both lie in } X \}.$$

X is closed under an ordering system if and only if every pair's σ lies in that pair's $C(x, y)$. A **pair constraint** at arity k is any relation $C \subseteq \{0, 1\}^k$ that is invariant under global complement $\sigma \mapsto \bar{\sigma}$ and contains 0^k and 1^k ; Lemma 6 shows every $C(x, y)$ is one and Theorem 7 that every one is some $C(x, y)$. A **partition constraint** is the pair constraint of a partition of $\{1, \dots, k\}$: the set of σ constant on every block.

D10 (bijunctive, majority). The ternary **majority** operation on $\{0,1\}$ is $\text{maj}(a,b,c) = (a \wedge b) \vee (b \wedge c) \vee (a \wedge c)$, extended coordinatewise to $\{0,1\}^m$. A relation $R \subseteq \{0,1\}^m$ is **bijunctive (majority-closed)** if $\text{maj}(a,b,c) \in R$ whenever $a,b,c \in R$. A relation over $\{0,1\}$ is expressible as a conjunction of clauses of at most two literals if and only if it is bijunctive — Schaefer (1978), and the general closure-property form is Jeavons, Cohen and Gyssens (1997), CITED. Bijunctivity is not invariant under an exclusive-or change of variables: the relation $\{(a, b, a \oplus b)\}$ is not majority-closed (the majority of 011, 101, 110 is 111), and §6.2 turns on that.

D11 (the growth step). For a reorderable Y with $|Y| \geq 2$, let $\text{drop}(Y)$ be the least $k \geq 1$ such that some subset of Y of size $|Y| - k$ is reorderable. For a box B , the **growth step** $\text{step}(B)$ is the maximum of $\text{drop}(Y)$ over all reorderable $Y \subseteq B$ with $|Y| \geq 2$. It is the largest jump that a procedure growing the reorderable family one cell at a time must be prepared to take; $\text{drop}(B) \leq \text{step}(B)$ always.

Notation. d is the number of coordinates; $x \leq y$ between tuples means $x_i \leq y_i$ for all i . The **median** of three cells is $m(x,y,z) = (x \wedge y) \vee (y \wedge z) \vee (z \wedge x)$, taken coordinatewise. Two tuples are at **Hamming distance** k when they differ in exactly k coordinates. In a closed X , a cell a is **join-irreducible** if it is not the join of two cells strictly below it — the bottom cell, having none, is counted here, which is a declared convention and not the standard one — and **join-prime** if $a \leq x \vee y$ with $x, y \in X$ implies $a \leq x$ or $a \leq y$; **meet-irreducible** and **meet-prime** are the duals, the top cell counted. A sublattice of a product of chains is distributive, and in a distributive lattice the irreducible and the prime elements coincide (Birkhoff 1967, CITED). Λ denotes the electron-configuration index of §5, and $|A_i|$ its alphabet sizes.

§2 · Closure, the defect, and the may-precede relation

2.1 Closure is what the envelopes express

Theorem 1 (the closure equivalence). Let X be an index with a fixed ordering system. Then

$$| \quad E(X) = 0 \iff X \text{ is closed.}$$

Moreover, when X is closed, $\mathcal{R}(X)$ is exactly the set of ambient points whose every pair-projection lies in $\pi_{ij}(X)$.

Proof. ($\Rightarrow$). Suppose $E(X) = 0$. With $X \subseteq \mathcal{R}(X)$ (D4) this gives $\mathcal{R}(X) = X$. Take $x, y \in X$. Both $x \wedge y$ and $x \vee y$ are ambient points, their coordinates being coordinates of cells. Fix $i \neq j$; it suffices to show $(x \vee y)_i \leq \varphi_{ij}((x \vee y)_j)$ and $(x \wedge y)_i \leq \varphi_{ij}((x \wedge y)_j)$.

For the join, $(x \vee y)_i$ is x_i or y_i ; say x_i . Then $(x \vee y)_j \geq x_j$, and φ_{ij} is non-decreasing because the defining set $\{z : z_j \leq v\}$ grows with v . Hence $\varphi_{ij}((x \vee y)_j) \geq \varphi_{ij}(x_j) \geq x_i$, the last step because $x \in X \subseteq \mathcal{R}(X)$.

For the meet, write $a = (x \wedge y)_i$ and $b = (x \wedge y)_j$. Four cases. If $a = x_i$ and $b = x_j$, then $a \leq \varphi_{ij}(x_j) = \varphi_{ij}(b)$. If $a = y_i$ and $b = y_j$, the same with y . If $a = x_i$ and $b = y_j$, then $a = x_i \leq y_i$ (that is what $a = x_i$ means when a is the smaller) and $y_i \leq \varphi_{ij}(y_j) = \varphi_{ij}(b)$. If $a = y_i$ and $b = x_j$, symmetrically $a = y_i \leq x_i \leq \varphi_{ij}(x_j) = \varphi_{ij}(b)$. In every case $x \wedge y \in \mathcal{R}(X) = X$.

($\Leftarrow$). Suppose X is closed. *Step 1: each pair projection is closed.* For $x, y \in X$, $\pi_{ij}(x \wedge y) = \pi_{ij}(x) \wedge \pi_{ij}(y)$ because meets are coordinatewise and projection only deletes coordinates; $x \wedge y \in X$, so the meet of any two elements of $\pi_{ij}(X)$ is in $\pi_{ij}(X)$, and likewise for joins.

Step 2: at two coordinates a closed set equals its own envelope region. Let $Y \subseteq A_0 \times A_1$ be closed with both alphabets fully used. By Theorem 3 below — whose proof uses nothing from this one — every fibre $F(u)$ is an interval $[\text{lo}(u), \text{hi}(u)]$ and lo, hi are non-decreasing. Then

$$| \quad \varphi_{10}(v) = \max \{ b : \exists a \leq v, (a,b) \in Y \} = \max \text{ over } a \leq v \text{ of } \text{hi}(a) = \text{hi}(v),$$

so the constraint $x_1 \leq \varphi_{10}(x_0)$ reads $x_1 \leq \text{hi}(x_0)$. And $\varphi_{01}(v) = \max \{ a : \text{lo}(a) \leq v \}$: since lo is non-decreasing the set $\{ a : \text{lo}(a) \leq v \}$ is a down-set of the chain A_0 , and it is non-empty because v is the second coordinate of some cell (a, v) , whose $\text{lo}(a) \leq v$; so $x_0 \leq \varphi_{01}(x_1)$ holds exactly when $\text{lo}(x_0) \leq x_1$. Therefore

$$\mid \mathcal{R}_2(Y) = \{ (a, b) : \text{lo}(a) \leq b \leq \text{hi}(a) \} = Y,$$

the last equality because the fibres are intervals.

Step 3: $\mathcal{R}$ is the intersection of its pair restrictions. By definition φ_{ij} depends on X only through $\pi_{ij}(X)$, and $A_i = A_i(\pi_{ij}(X))$. So

$$\mid \mathcal{R}(X) = \{ x : (x_i, x_j) \in \mathcal{R}_2(\pi_{ij}(X)) \text{ for all } i < j \},$$

and by Steps 1 and 2 each $\mathcal{R}_2(\pi_{ij}(X)) = \pi_{ij}(X)$. Hence $\mathcal{R}(X)$ is the set of ambient points all of whose pair-projections are projections of cells.

Step 4: a sublattice is determined by its pair projections. Each chain A_i , as a lattice, satisfies the majority identities for the median term $m(x, y, z) = (x \wedge y) \vee (y \wedge z) \vee (z \wedge x)$: $m(x, x, y) = x \vee (x \wedge y) = x$ by absorption, and the other two arguments by symmetry of the expression. X is a sublattice of $\prod_i A_i$ and hence closed under m . By the theorem of Baker and Pixley (1975), CITED — an algebra with a majority term has the property that every subalgebra of a finite product is the intersection of the cylinders over its two-fold projections — X equals that intersection. Combining with Step 3, $\mathcal{R}(X) = X$, so $E(X) = 0$. ■

EXHAUSTIVE. The equivalence, and the identification of $\mathcal{R}(X)$ with the pair-projection intersection when $E = 0$, were checked on **every** subset of the boxes 3×3 , 2×4 , $2 \times 2 \times 2$, $2 \times 2 \times 3$ and $2 \times 2 \times 2 \times 2$ with at least two cells — 502, 247, 247, 4,083 and 65,519 subsets, **70,598** in all, no disagreement and no failure. The envelope computation was checked cell for cell against an independent implementation on 120 further instances (§8, guard (d)).

Corollary 1 (self-expression, and totality). If X is closed then

$$\mid X = \{ x \in \prod_i A_i : x_i \leq \varphi_{ij}(x_j) \text{ for all } i \neq j \},$$

so the alphabets and the $d(d-1)$ envelopes determine X completely; and the membership predicate of X is a conjunction of $d(d-1)$ comparisons between elements of finite chains, hence **decides every ambient point**. There is no ambient point on which membership is undetermined.

Proof. The display is Theorem 1. Each comparison $x_i \leq \varphi_{ij}(x_j)$ is a comparison of two elements of the chain A_i , decidable; a finite conjunction of decidable predicates is decidable, and its domain is the whole ambient box. ■

PROVED, and **EXHAUSTIVE** for Λ : the conjunction of the 56 envelope comparisons was evaluated at all **6,912** ambient points, returned a value at every one, and agreed with membership at every one; so was the conjunction of Λ 's seven defining bounds (§5.1, §7.1).

Corollary 1 is the reason the remaining questions are about the *order* and not about the alphabet or the bounds. The alphabet is a projection and always available; the bounds are a maximum over cells and always available; only the order is a choice that the cells might or might not determine.

2.2 The necessity of “may precede”

Lemma 1 (necessity, and it is dimension-free). Let X be closed under an ordering system s , fix a coordinate i , and let u precede v in s_i . Then $u \sqsubseteq v$.

Proof. Let $p \in F(u)$ and $q \in F(v)$, and let x and y be the cells with $x_i = u$, $x_{-i} = p$ and $y_i = v$, $y_{-i} = q$. Then $x \wedge_s y$ has i -th coordinate the s_i -smaller of u and v , which is u , and its remaining coordinates are $p \wedge q$. Since X is closed, $x \wedge_s y \in X$, so $p \wedge q \in F(u)$. Likewise $x \vee_s y$ has i -th coordinate v and remaining coordinates $p \vee q$, so $p \vee q \in F(v)$. ■

Nothing in the argument uses $d = 2$, and nothing uses any property of the other coordinates beyond their being ordered.

Lemma 2 (reflexivity and totality on a closed index). If X is closed under s then, on every coordinate, $\sqsubseteq$ is reflexive and total.

Proof. Reflexivity: for $p, q \in F(u)$, the cells over u with those tails have meet with i -th coordinate u and tail $p \wedge q$, and join with i -th coordinate u and tail $p \vee q$; both lie in X by closure, so both tails lie in $F(u)$. Totality: s_i is a linear order, so for $u \neq v$ one of the two precedes the other in s_i , and Lemma 1 applies. ■

Lemma 3 (transitivity, with no hypothesis). For every index X , every coordinate i , and every ordering of the remaining coordinates: if $u \sqsubseteq v$ and $v \sqsubseteq w$ then $u \sqsubseteq w$.

Proof. Let $p \in F(u)$ and $r \in F(w)$. By D1 the fibre $F(v)$ is non-empty; fix $q \in F(v)$.

The meet. From $u \sqsubseteq v$ applied to p and q : $p \vee q \in F(v)$. From $v \sqsubseteq w$ applied to $p \vee q \in F(v)$ and $r \in F(w)$: $(p \vee q) \wedge r \in F(v)$. From $u \sqsubseteq v$ applied to $p \in F(u)$ and $(p \vee q) \wedge r \in F(v)$:

$$\mid p \wedge ((p \vee q) \wedge r) \in F(u).$$

By associativity and the absorption law $p \wedge (p \vee q) = p$, the left side is $p \wedge r$. So $p \wedge r \in F(u)$.

The join. From $v \sqsubseteq w$ applied to q and r : $q \wedge r \in F(v)$. From $u \sqsubseteq v$ applied to $p \in F(u)$ and $q \wedge r \in F(v)$: $p \vee (q \wedge r) \in F(v)$. From $v \sqsubseteq w$ applied to $p \vee (q \wedge r) \in F(v)$ and $r \in F(w)$:

$$\mid (p \vee (q \wedge r)) \vee r \in F(w),$$

and by associativity and absorption $(q \wedge r) \vee r = r$, so the left side is $p \vee r$. So $p \vee r \in F(w)$. Both requirements of $u \sqsubseteq w$ hold. ■

Lemma 3 is the reason the criterion below is a criterion and not merely a necessary condition: the relation an index hands over is always transitive, so the only two things that can fail are reflexivity and totality. Computing $\sqsubseteq$ costs one meet, one join and two membership tests for every pair (p, q) with $p \in F(u)$, $q \in F(v)$, over all u, v — that is $\sum_{u,v} |F(u)| \cdot |F(v)| = |X|^2$ pairs, $O(d \cdot |X|^2)$ operations — and the two tests are then $O(|A_i|^2)$.

2.3 The criterion on one coordinate

Theorem 2 (one-coordinate order recovery). Let X be an index, fix a coordinate i , order every other coordinate, and let $\sqsubseteq$ be the resulting relation on A_i .

(a) $\sqsubseteq$ is transitive. (b) For a linear order s on A_i : X is closed under $(s, \text{the rest})$ if and only if $\sqsubseteq$ is reflexive and $u <_s v \Rightarrow u \sqsubseteq v$. (c) X is closed under *some* order on A_i if and only if $\sqsubseteq$ is a total preorder; and in that case the admissible orders on A_i are exactly the linear extensions of $\sqsubseteq$.

Proof. (a) is Lemma 3.

1. ($\Rightarrow$) is Lemmas 1 and 2. ($\Leftarrow$) Suppose $\sqsubseteq$ is reflexive and $s \subseteq \sqsubseteq$. Take cells $x = (u, p)$ and $y = (v, q)$, written with the i -th coordinate first. If $u = v$, reflexivity gives $p \wedge q, p \vee q \in F(u)$, so both $x \wedge y = (u, p \wedge q)$ and $x \vee y = (u, p \vee q)$ are cells. If $u <_s v$, then $u \sqsubseteq v$, so $p \wedge q \in F(u)$ and $p \vee q \in F(v)$; and $x \wedge y = (u, p \wedge q)$ and $x \vee y = (v, p \vee q)$, both cells. The case $v <_s u$ is the same with the roles exchanged. So X is closed.
2. ($\Rightarrow$) If X is closed under some s , then by (b) $\sqsubseteq$ is reflexive and contains s ; s is total, so $\sqsubseteq$ is total; and $\sqsubseteq$ is transitive by (a). ($\Leftarrow$) Suppose $\sqsubseteq$ is a total preorder. Let s be any linear extension. By (b) it suffices to show $s \subseteq \sqsubseteq$, which is the definition of a linear extension, and that at least one exists — Lemma 4. Conversely, if X is closed under s then $s \subseteq \sqsubseteq$ by (b), so s is a linear extension. ■

Lemma 4 (the recovered order is a linear extension, and the extensions are the answer). Let $\sqsubseteq$ be a total pre-order on a finite set A , and let $\text{rank}(u) = |\{v : v < u\}|$. Then sorting A by $(\text{rank}(u), u)$ — the recovered order s^* of D7 — is a linear order and a linear extension of $\sqsubseteq$. The set of linear extensions of $\sqsubseteq$ is non-empty, and its size is the product, over the $\sqsubseteq$ -equivalence classes c , of $|c|!$.

Proof. s^* is a sort by a key with distinct second components, hence a strict linear order. Suppose $u <_{s^*} v$ but not $u \sqsubseteq v$. By totality $v \sqsubseteq u$, and with “not $u \sqsubseteq v$ ” this gives $v < u$. Every w with $w < v$ satisfies $w < u$: $w \sqsubseteq v \sqsubseteq u$ gives $w \sqsubseteq u$ by transitivity, and $u \sqsubseteq w$ would give $u \sqsubseteq v$ by transitivity, which is excluded. Also $v < u$ while not $v < v$ ($\sqsubseteq$ is reflexive). So $\{w : w < v\} \subsetneq \{w : w < u\}$ and $\text{rank}(v) < \text{rank}(u)$, contradicting $u <_{s^*} v$, which requires $\text{rank}(u) \leq \text{rank}(v)$. So s^* is a linear extension, and the set is non-empty.

For the count: $u \sqsubseteq v$ and $v \sqsubseteq u$ is an equivalence (reflexive by hypothesis, symmetric by construction, transitive by (a)), and $\sqsubseteq$ induces a linear order on the classes, by totality and transitivity. A linear extension must order the classes that way — if c precedes c' and $u \in c$, $v \in c'$, then $v \sqsubseteq u$ fails, so $u <_s v$ by the extension property applied to s total — and may order each class internally in any of $|c|!$ ways, every such choice giving an extension. ■ **PROVED.**

MACHINE-CHECKED. Theorem 2 (a), (b) in both directions, and (c) in the form “ $\sqsubseteq$ reflexive and total $\Rightarrow s^*$ is a linear order and X is closed under s^* ”, were put to Z3 with X a vector of Booleans ranging over **every** subset of the box that uses every value and the order on the first coordinate a vector of Booleans constrained to be antisymmetric, total and transitive — so the solver quantifies over subsets and orderings together. Four boxes: 3×3 (2^9 subsets $\times$ 6 orders), 4×4 ($2^{16} \times 24$), $2 \times 2 \times 3$ ($2^{12} \times 2$) and $3 \times 3 \times 3$ ($2^{27} \times 6$). Sixteen obligations, all unsat on the negation. Both guards were run first and passed (§8).

REFUTATION ($\sqsubseteq$ need not be antisymmetric). “Closed under some order implies $\sqsubseteq$ is antisymmetric” is false. Z3 produced a witness in each of the four boxes; the simplest is the **full box** itself, which is closed under every ordering system and has every fibre equal to everything, so $u \sqsubseteq v$ for all u, v . This is not a defect of the criterion but its content: ties in $\sqsubseteq$ are exactly the values the cells do not distinguish, and Lemma 4 counts the resulting ambiguity.

2.4 Why the alphabet convention is load-bearing

If A_1 were allowed to contain a value no cell uses, the fibre over it would be empty, and $u \sqsubseteq v$ would hold vacuously for that v against every u and in the other direction too — the relation would be a total preorder with a spurious class, and the recovered order would be an order on values the index does not carry. The convention $A_i = \{x_i : x \in X\}$ removes that. It is also what makes Lemma 3 available: the proof picks an element of $F(v)$, and there is one only because v is used. In Theorem 3 the convention does more work still, and §3.1 exhibits a closed set whose fibre is not an interval of a declared alphabet.

§3 · Two coordinates: a complete criterion

3.1 The fibre characterisation

Theorem 3 (fibres at $d = 2$). Let $X \subseteq A_0 \times A_1$ with A_1 ordered and every value of both alphabets used, and let s be an order on A_0 . Then X is closed under (s, A_1) if and only if

1. every fibre $F(u)$ is an interval of the chain A_1 , and
2. lo and hi are both non-decreasing along s .

Proof. ($\Rightarrow$). Intervals. Let $b_1 < b_2 < b_3$ in A_1 with $(u, b_1), (u, b_3) \in X$, and suppose $(u, b_2) \notin X$. By D1 the value b_2 is used, so there is $u' \neq u$ with $(u', b_2) \in X$. If $u' <_s u$, then $(u', b_2) \vee (u, b_1) = (u, \max(b_2, b_1)) = (u, b_2)$, which closure puts in X — contradiction. If $u <_s u'$, then $(u, b_3) \wedge (u', b_2) = (u, \min(b_3, b_2)) = (u, b_2)$ — contradiction again. So $(u, b_2) \in X$ and $F(u)$ is an interval.

Monotone endpoints. Let $u <_s v$. Applying closure to $(u, \text{lo}(u))$ and $(v, \text{lo}(v))$, the meet is $(u, \min(\text{lo}(u), \text{lo}(v))) \in X$, so $\min(\text{lo}(u), \text{lo}(v)) \geq \text{lo}(u)$, that is $\text{lo}(v) \geq \text{lo}(u)$. Applying it to $(u, \text{hi}(u))$ and $(v, \text{hi}(v))$, the join is $(v, \max(\text{hi}(u), \text{hi}(v))) \in X$, so $\max(\text{hi}(u), \text{hi}(v)) \leq \text{hi}(v)$, that is $\text{hi}(u) \leq \text{hi}(v)$.

($\Leftarrow$). Let $x = (u, b)$ and $y = (v, c)$ be cells with $u \leq_s v$. The meet is $(u, \min(b, c))$. Now $\min(b, c) \leq b \leq \text{hi}(u)$; and $\min(b, c) \geq \text{lo}(u)$ because $b \geq \text{lo}(u)$ and $c \geq \text{lo}(v) \geq \text{lo}(u)$. Since $F(u)$ is the interval $[\text{lo}(u), \text{hi}(u)]$, the meet is a cell. The join is $(v, \max(b, c))$, and $\max(b, c) \geq c \geq \text{lo}(v)$, while $\max(b, c) \leq \text{hi}(v)$ because $c \leq \text{hi}(v)$ and $b \leq \text{hi}(u) \leq \text{hi}(v)$. So the join is a cell. ■

MACHINE-CHECKED. The biconditional was put to Z3 with X ranging over every subset of the box that uses every value and s over every total order on A_0 , in four boxes: 3×3 ($2^9 \times 6$), 4×4 ($2^{16} \times 24$), 3×5 ($2^{15} \times 6$) and 5×4 ($2^{20} \times 120$). Four obligations, all *unsat* on the negation, both guards passed.

The convention of D1 is used exactly once in the proof, in the interval half, and it cannot be dropped. **REFUTATION, with the witness printed.** Declare the second alphabet to be $\{0,1,2\}$ and take $X' = \{(0,0), (0,2), (1,0), (1,2)\}$. Then $X' = \{0,1\} \times \{0,2\}$ is closed under both orders on A_0 , while its fibre over $u = 0$ is $\{0,2\}$, which is not an interval of the declared chain $\{0,1,2\}$: the theorem fails for a declared alphabet. Under D1 the second alphabet of X' is $\{0,2\}$, of which $\{0,2\}$ is an interval, and the theorem holds. Adding the cell $(1,1)$ makes the value 1 used, and then the meet of $(0,2)$ and $(1,1)$ is $(0,1)$, which is not a cell — the set with $(1,1)$ is not closed, exactly as the theorem says.

3.2 The nesting lemma

Say the interval I **nests strictly** inside J when $\text{lo}(J) < \text{lo}(I)$ and $\text{hi}(I) < \text{hi}(J)$.

Lemma 5 (nesting). A finite family S of intervals of a chain — a multiset, since two fibres may coincide, and a coincidence is a tie of $\sqsubseteq$ — admits an ordering in which both endpoints are non-decreasing if and only if no member of S nests strictly inside another. Sorting by left endpoint, ties by right endpoint, produces such an ordering whenever one exists.

Proof. ($\Rightarrow$) Suppose I nests strictly inside J . In any ordering either I precedes J , which requires $\text{lo}(I) \leq \text{lo}(J)$ — false — or J precedes I , which requires $\text{hi}(J) \leq \text{hi}(I)$ — also false.

($\Leftarrow$) Sort by (lo, hi) . Left endpoints are non-decreasing by construction. Suppose some consecutive pair I then J has $\text{hi}(J) < \text{hi}(I)$. The sort gives $\text{lo}(I) \leq \text{lo}(J)$. If $\text{lo}(I) < \text{lo}(J)$ then J nests strictly inside I , contrary to hypothesis. If $\text{lo}(I) = \text{lo}(J)$ then the tie-break orders by hi , so $\text{hi}(I) \leq \text{hi}(J)$, contradicting $\text{hi}(J) < \text{hi}(I)$. So no right endpoint decreases, and the sorted order is monotone in both. Nothing in either direction uses that the members are distinct. ■

EXHAUSTIVE. Over **every** set of two, three or four distinct intervals of a ground chain of two, three, four or five points — **2,354** families — and over every multiset of two, three or four intervals of those chains, repeats allowed — **5,084** families — the existence of a monotone ordering, the absence of strict nesting, and the verdict of the sort test all agree, with no exception.

REFUTATION (the qualifier is not decoration). Reading “nests” as ordinary containment — $I \subseteq J$, $I \neq J$ — makes the lemma false on **1,098** of the 2,354 families of distinct intervals. An interval sharing an endpoint with its container does not nest strictly, and the family $\{ [0,1], [0,2] \}$ is monotone in the order given while being nested in the weak sense.

The families of Lemma 5 are the interval systems of the consecutive-ones literature. What a PQ-tree decides — Booth and Lueker (1976), Tucker (1972), CITED — is whether the rows of the fibre matrix can be permuted so that every column's ones are consecutive, which is the one-coordinate problem with the order of the *other* coordinate free. Lemma 5 asks the narrower question, with the second order fixed, and one sort settles it.

3.3 The decision at two coordinates, with both orders unknown

Theorem 3 and **Lemma 5** together decide reorderability at $d = 2$ when one order is fixed: the fibres must be intervals, and no fibre may nest strictly inside another; the sort then produces the order. When **both** orders are unknown, one enumerates the orders of the smaller alphabet and applies Theorem 2 to the other.

DECIDE₂(X). For each of the $|A_0|!$ orders s_0 of A_0 : compute $\sqsubseteq$ on $A_1 - |X|^2$ meets and joins, by §2.2 — and test whether it is reflexive and total. If it is, return s_0 and the recovered order s^* on A_1 . If no s_0 succeeds, report that X is not reorderable.

Correctness is Theorem 2(c) applied to the coordinate 1 for a fixed s_0 , together with the observation that closure under (s_0, s_1) for some s_1 is exactly what the inner test decides. The cost is $|A_0|!$ times $O(|X|^2)$.

EXHAUSTIVE. DECIDE₂ was compared against brute force — closure tested under all $|A_0|! \cdot |A_1|!$ ordering systems — on **every** subset of five boxes that uses every value of both alphabets: 2×3 (25 subsets), 2×4 (79), 3×3 (265), 3×4 (2,161) and 4×4 (41,503). **44,033** instances, agreement on every one.

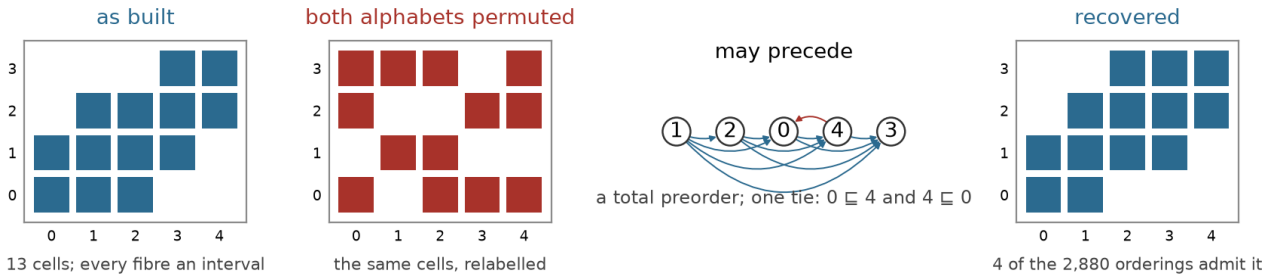


Figure 1. A 13-cell index in the 5×4 box, built with $lo = (0,0,0,1,2)$ and $hi = (1,2,2,3,3)$, then relabelled on both coordinates, then recovered. Left: as built; every fibre is an interval and both endpoints are non-decreasing (Theorem 3). Centre left: the same thirteen cells after both alphabets are independently permuted. Centre right: the may-precede relation on the first coordinate at the recovered order of the second — a total preorder with one tie, drawn as the one pair of opposed arrows. Right: recovered — the ordering system returned is the reverse of the one the index was built with, which is admissible by D3 and is what the cells determine. Four of the 2,880 ordering systems admit this index: the tie contributes a factor of two and reversing every coordinate contributes the other.

§4 · Above two coordinates

4.1 Pointwise is not enough

Every pair projection of a reorderable index is reorderable (Theorem 1, Step 1, applied under the admissible ordering system). The converse fails, and the failure is not an artefact of any criterion: closure is a d -dimensional condition, and Theorem 1 Step 4 recovers a *closed* set from its pair projections exactly because it is closed.

REFUTATION, with the witness proved. Let $X = \{ 100, 010, 001 \} \subseteq \{0,1\}^3$, the three unit vectors. Every value of every coordinate is used. Each pair projection is a three-element subset of $\{0,1\}^2$, and every such subset is reorderable: flipping one coordinate if necessary carries the omitted point to 01, and $\{ 00, 10, 11 \} = \{ x : x_1 \leq x_0 \}$ is a sublattice. Yet X is closed under no ordering system.

Proof. An ordering of a two-value alphabet is the identity or the transposition, so an ordering system on $\{0,1\}^3$ is a pattern of coordinate flips, and it carries X to three cells x' , y' , z' that are still pairwise at Hamming distance two. Take x' and y' . They agree on exactly one coordinate, c , and differ on the other two, so their meet and join are the two remaining vertices of the square on which they differ: both distinct from x' and y' , distinct from each other, and both agreeing with x' and y' on c . The third cell z' differs from x' on c — before the flips, z is the unit vector supported on c — so neither the meet nor the join is z' . Closure would require two cells outside X . ■

EXHAUSTIVE. Over every subset of three boxes that uses every value of every alphabet — **193** of $2 \times 2 \times 2$, **3,271** of $2 \times 2 \times 3$ and **63,775** of $2 \times 2 \times 2 \times 2$ — every pair projection of every one is reorderable, while **98, 2,460** and **61,462** of them respectively are not. In these boxes the pair criterion excludes nothing at all, because every subset of a 2×2 or a 2×3 box is reorderable (Table 2: 16 of 16 and 64 of 64), and the smallest non-reorderable subsets the census finds are the three unit vectors above, $\{001, 002, 010, 100\}$ in $2 \times 2 \times 3$, and $\{0011, 0100, 1000\}$ in $2 \times 2 \times 2 \times 2$. What restores the pointwise reading is a hypothesis on the constraint graph, and Theorem 4 states it; §4.4 measures how the failure splits when the hypothesis is absent.

4.2 Graph structure, and why it restores the pointwise reading

Theorem 4 (factorisation over a graph). Let G be a graph on $\{1, \dots, d\}$ and let X be G -structured (D8). Then, for any ordering system s ,

$$X \text{ is closed under } s \iff \text{for every edge } (i,j) \text{ of } G, \pi_{ij}(X) \text{ is closed under } (s_i, s_j).$$

Proof. ($\Rightarrow$) is Step 1 of Theorem 1: projections of closed sets are closed.

($\Leftarrow$) Let $x, y \in X$ and let $z = x \wedge y$. For each edge (i,j) , (x_i, x_j) and (y_i, y_j) lie in $\pi_{ij}(X)$, so their meet — which is exactly (z_i, z_j) , meets being coordinatewise — lies in $\pi_{ij}(X)$ by hypothesis. Since X is the set of ambient points satisfying every edge condition, $z \in X$. The join is identical with v . ■

Nothing in the proof uses that G is a tree: for a G -structured index the joint condition on the d orders is a conjunction of conditions on *pairs* of orders, one per edge of G . What a tree adds is that the conjunction can be satisfied by a sweep, because a tree has no cycles; that is what Theorem 5 needs, and it is the backtrack-free regime of Freuder (1982), CITED, the width-one case of the tree-clustering and local-to-global results of Dechter and Pearl (1989) and Dechter (1992), CITED. For monotone binary constraints — the closed case of D4 — Montanari (1974), CITED, shows that path consistency already implies global consistency.

4.3 The recovery algorithm

RECOVER(X, T, r): root T at r ; for each of the $|A_r|!$ linear orders s_r of A_r , output EXTEND(r, s_r).

EXTEND(i, s_i): the set of ordering systems of the subtree at i that extend s_i , memoised on the pair (i, s_i)
 for each child j of i :
 compute $\sqsubseteq$ on A_j from $\pi_{ij}(X)$, with A_i ordered by s_i
 if $\sqsubseteq$ is not a total preorder: return $\emptyset$
 $S_j \leftarrow \bigcup \{ \text{EXTEND}(j, s_j) : s_j \text{ a linear extension of } \sqsubseteq \}$; if $S_j = \emptyset$: return $\emptyset$
 return $\{ s_i \text{ together with one choice from each } S_j \}$

Theorem 5 (correctness, and the cost). For a T -structured X , RECOVER returns exactly the ordering systems under which X is closed — every one of them, and nothing else. Let $F = \sum_i |A_i|!$. With EXTEND memoised on (i, s_i) , deciding whether X is reorderable and returning one admissible ordering system costs $O(F \cdot (d \cdot |X|^2 + F))$ operations.

Proof. Correctness. By Theorem 4 the closure condition is the conjunction over the edges of T of a condition on (s_i, s_j) . Root T at r . The edges of the subtree at a node i involve only the coordinates of that subtree, and the subtrees at the children of i are vertex-disjoint and share no edge; the only edge joining the subtree at a child j to the rest is (i, j) . Therefore, once s_i is fixed, the condition splits as a product over the children: an ordering system of the subtree at i extending s_i is admissible if and only if, for each child j , the pair (s_i, s_j) satisfies the edge condition and the restriction to the subtree at j is admissible there. That is exactly the recursion **EXTEND** computes.

For one edge (i, j) with s_i fixed, $\pi_{ij}(X)$ is a two-coordinate index with the first coordinate ordered, and Theorem 2 says: an s_j closing it exists exactly when $\sqsubseteq$ on A_j is a total preorder, and the closing orders are exactly the linear extensions of $\sqsubseteq$. That is the inner loop. Since the recursion enumerates every extension at every node and every order at the root, and rejects nothing that satisfies every edge condition, the returned set is exactly the admissible set.

Cost. The memo key (i, s_i) takes at most $|A_i|!$ values at node i , so **EXTEND** is evaluated at most F times in all; every other call is a lookup. One evaluation at i does, for each child j : compute $\sqsubseteq$ on A_j from $\pi_{ij}(X)$, which by §2.2 is $|\pi_{ij}(X)|^2 \leq |X|^2$ meets and joins of pairs, $O(|X|^2)$; test reflexivity and totality, $O(|A_j|^2) \leq O(|X|^2)$; and enumerate the linear extensions of $\sqsubseteq$, at most $|A_j|!$ of them, each a lookup or an evaluation charged to its own key. A node has fewer than d children, so an evaluation costs $O(d \cdot |X|^2)$ plus at most $\sum_j |A_j|! \leq F$ lookups. Summing over at most F evaluations: $O(F \cdot d \cdot |X|^2 + F^2)$. In the decision form each child's enumeration stops at the first extension whose subtree is feasible, and the returned system is assembled once. ■ **PROVED.**

In the form that returns every solution, the sets S_j are formed explicitly and their products taken; a subtree's solutions may be assembled at an evaluation that a sibling later rejects, so no bound in terms of the size of the output is claimed. What the memoisation buys is $F = \sum_i |A_i|!$ in place of $\prod_i |A_i|!$: on Λ , 94 against 11,943,936 (§5.3).

EXHAUSTIVE. RECOVER was compared with brute force on **every** T -structured subset that uses every value of every alphabet, in four families, the solution sets compared and not merely their emptiness:

box	tree	T-structured subsets	solution sets equal	of these, re-orderable	relabellings tested	recovered
$2 \times 2 \times 3$	path 0–1–2	175	175	175	4,200	4,200
$2 \times 3 \times 3$	path 0–1–2	6,625	6,625	4,819	346,968	346,968
$2 \times 2 \times 2 \times 2$	path 0–1–2–3	343	343	343	5,488	5,488
$2 \times 2 \times 2 \times 2$	star 0–{1,2,3}	343	343	343	5,488	5,488

Table 1. Every T -structured subset of each box; for each, the set of ordering systems RECOVER returns against the set brute force finds. The last two columns take each reorderable subset and each of the $\prod |A_i|!$ relabellings of it, and ask RECOVER to recover an admissible order from the relabelled bag: **362,144** scrambles, **362,144** recovered. That 4,819 of the 6,625 T -structured subsets of the $2 \times 3 \times 3$ box are reorderable — and 1,806 are not — is the measurement that tree structure is not by itself closability.

4.4 The defect diagnosed

Theorem 1 says a defect exists exactly when the index is not closed. The diagnostic question is what *kind* of failure it is, and the answer is a dichotomy with a witness on each side.

Theorem 6 (the diagnostic split). For an index X with a fixed ordering system, $E(X) > 0$ if and only if at least one of the following holds: **(A)** some pair projection $\pi_{ij}(X)$ is not closed in $A_i \times A_j$; **(B)** some three cells $x, y, z \in X$ have $m(x, y, z) \notin X$.

Proof. If $E(X) = 0$ then X is closed (Theorem 1), so every projection is closed (Step 1) and X is closed under the median, which is built from meets and joins. So neither (A) nor (B).

Conversely suppose neither (A) nor (B). Since every $\pi_{ij}(X)$ is closed, Steps 2 and 3 of Theorem 1 give $\mathcal{R}(X) = \{x : (x_i, x_j) \in \pi_{ij}(X) \text{ for all } i < j\}$. Since (B) fails, X is closed under the median, hence is a subalgebra for a majority term, hence by Baker and Pixley (1975) equals the intersection of the cylinders over its pair projections. The two coincide, so $\mathcal{R}(X) = X$ and $E(X) = 0$. ■ **PROVED.**

EXHAUSTIVE. Over the same 70,598 subsets of five boxes, every subset with $E(X) > 0$ carries a witness of type (A), of type (B), or of both; **none carries neither**. The split is informative rather than trivial: at the box $2 \times 2 \times 2$, of 182 defective subsets, 92 have only a pair fault, 52 only a median witness, 38 both; at $2 \times 2 \times 2 \times 2$, of 64,804 defective subsets, 3,438 have only a pair fault, 38,768 only a median witness, 22,598 both.

The diagnostic loop, as a procedure. Given an index and a coordinatisation:

1. Compute the envelopes and $E(X)$. If $E(X) = 0$, stop: Corollary 1 gives the alphabet, the bounds and a total membership test, and the index expresses itself.
2. Otherwise locate a witness by Theorem 6. A pair witness names two coordinates and four cells; a median witness names three cells and the ambient point the index refuses. **A witness is a missing constraint:** the ambient point it produces is admitted by every monotone binary bound the cells support and excluded by X , so whatever excludes it is a form the envelopes cannot carry.
3. A pair witness may be an artefact of the labelling. Decide by Theorem 2 whether that pair projection is closable under some order on one of its two coordinates; if it is, the recovered order removes that witness. A median witness, under the given ordering, is not removed by any operation that leaves every pair projection as it is — by Baker and Pixley it is precisely the obstruction to being determined by them. A re-ordering changes meets and joins and so may remove it; whether one does is the reorderability question of §6.
4. If no admitted operation on the coordinates reduces E , the excess is a statement about the subject and not about the drawing.

Step 3 is where the dichotomy of Freuder (1978), CITED, appears with witnesses attached: a failure that a re-ordering repairs, against a failure of *arity* that names more coordinates than a pairwise bound has arguments.

A worked instance. Take the periodic system. Coordinatised by (period, group) in the eighteen-column layout (Scerri 2007, CITED), the cells are elements: the occupied cells number **90** in a box of 126, and $E = 36$ — the closure admits every ambient point, which is to say the two coordinates support no monotone bound at all. Coordinatised instead by $(n + \ell, \ell)$ — the left-step arrangement of Janet (1928), CITED — the cells are subshells, not elements: the **19** subshells 1s to 7p that are occupied through $Z = 118$ sit in a box of 32, and $E = 0$; with 8s added, **20** cells, and again $E = 0$. The defect of the first is not a property of the elements. It is a property of the drawing, and re-coordinatisation removes it. That the two indexes have different cells — elements in one, subshells in the other — is the point: a re-coordinatisation is not a bijection of cells, and the operation admitted in step 4 is a change of what a cell is.

§5 · Λ , and what an index recovers about itself

5.1 The index

Λ indexes transitions between electron configurations on eight coordinates

$$\mid (n, \ell, k, q, e, f, g, 2S)$$

— source shell, source subshell, source occupancy, electrons removed, target shell, target subshell, target occupancy, and twice the total spin — subject to seven bounds:

$$\mid \ell \leq n - 1, k \leq 4\ell + 2, q \leq k, f \leq e - 1, g \leq 4f + 2, g \leq q, 2S \leq k.$$

The first and fourth are the range of ℓ for principal quantum number n , the second and fifth are the Pauli capacity of a subshell, the third and sixth are counting conventions of the index — g counts the moved electrons only, whatever the target subshell already holds — and the seventh is an envelope on the total spin, not a selection rule: it admits $2S$ of either parity for every k , so Λ holds the cell $(1, 0, 1, 0, 1, 0, 0, 0)$, one electron with $2S = 0$, and **413** of its 976 cells carry a $2S$ of the wrong parity for their k . The parity rule $2S \equiv k \pmod{2}$ is not of the form D4 admits and could not be a bound of the index. Λ is therefore an index of *admissible labels* under seven envelope bounds, not of realised configurations, and §5.3 and §7.3 are read with that in mind. Every one of the seven is of the form “one coordinate bounded by a non-decreasing function of one other”, which is what puts Λ inside the envelope class of D4. At the caps $n, e \leq 3, \ell, f \leq 1, k \leq 3$, Λ has **976** cells in a box of **6,912**, and $E(\Lambda) = 0$; at the caps $n, e \leq 2, k \leq 2$ it has **216**.

EXHAUSTIVE. All of: 976 cells, box 6,912, $E = 0$ under the envelope operator; the conjunction of the seven bounds evaluated at all 6,912 ambient points, returning a value at each and agreeing with membership at each, and the conjunction of the 56 envelope comparisons of Corollary 1 likewise; the parity count; and the reconstruction of all 976 cells from the seven bounds alone, at both cap settings.

5.2 Λ 's own alphabet and its constraint graph

Counting the bottom cell as join-prime and the top as meet-prime (§1, Notation), Λ has **18** cells that are join-prime and **18** that are meet-prime, and **no cell is both**. The 18 join-primes are the bottom cell and the 17 join-irreducibles above it, one for each value of each coordinate above that coordinate's minimum: $\sum_i (|A_i| - 1) = 2+1+2+3+2+1+3+3 = 17$. The envelope operator finds **16 binding coordinate pairs**, not 7; but Λ is T-structured for the seven-edge tree

$$| \quad n - \ell - k - q - g - f - e, \text{ with } 2S \text{ attached to } k,$$

so the nine further binding pairs are implied and nothing is lost. The tree is a caterpillar with one leg, and Λ is exactly the intersection of the cylinders over its seven edge projections — EXHAUSTIVE, checked over all 6,912 ambient points at the larger cap setting and over the whole box at the smaller.

5.3 Recovery from a scrambled bag

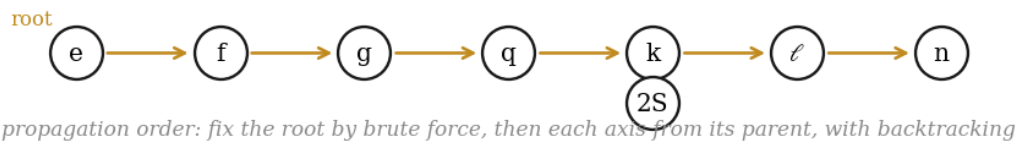
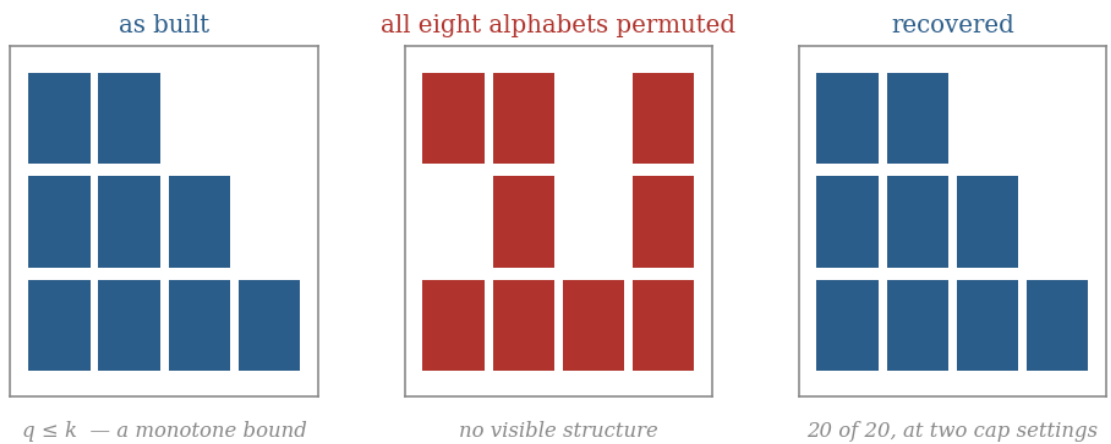


Figure 2. Two of Λ 's eight coordinates. Left: as built, showing the monotone bound $q \leq k$. Centre: the same cells after all eight alphabets are independently permuted — the structure is not visible. Right: recovered. Below, the propagation order along the constraint tree: the root is fixed by enumeration and each further coordinate is solved from its parent. The plate's phrase "with backtracking" refers to the enumeration of the root's orders and of the linear extensions at each node, which is what RECOVER does; there is no other search. Twenty of twenty scrambles recovered, at two cap settings.

SAMPLED, with the size and the seed stated. Twenty pseudorandom scrambles at each cap setting, seed 2026, every one of the eight alphabets independently permuted. RECOVER rooted at e returned an ordering system under which the scrambled bag is closed in **20 of 20** cases at 976 cells and **20 of 20** at 216 cells. At most 2 of the 6 root orders were tried before a solution was found.

EXHAUSTIVE, and it is the honest qualifier on the sample. Running RECOVER with every solution returned rather than the first, Λ is closed under exactly **16** ordering systems, at both cap settings — out of $\prod_i |A_i|! = 11,943,936$. The recovery therefore cannot be expected to return the ordering Λ was built with: it returned the built ordering or its reverse in **3 of 20** scrambles at 976 cells and **0 of 20** at 216. What is recovered is an admissible ordering, and *admissible* is all the cells determine. The correct statement of the result is: **the recovery succeeded 20 times out of 20 at each setting, and the answer is unique up to the 16 orderings the index admits.**

Which sixteen. The two settings' sixteen are different objects. At 976 cells they are one global reversal times three *ties* — pairs of values with identical fibres, which no cell distinguishes: $n: 2 \sim 3$ and $e: 2 \sim 3$, because with ℓ , $f \leq 1$ the shells 2 and 3 carry the same subshells in the index, and $2S: 0 \sim 1$, because $2S \leq k$ admits both for every k — the over-admission of §5.1, seen from the other side; every edge of the tree is binding, so the tree is one piece and reversal is all-or-nothing. At 216 cells the only tie is $2S: 0 \sim 1$, and the other factor of eight is the *independent reversal of three components*: with $k \leq 2$ the bounds $k \leq 4\ell + 2$ and $g \leq 4f + 2$ are vacuous, the pair projections onto the edges $\ell-k$ and $f-g$ are full products, and the tree falls into the pieces $\{n, \ell\}$, $\{e, f\}$ and $\{k, q, g, 2S\}$, each of which can be reversed on its own. D3's reverse pairs are the one-component case; at 216 a proper subset of the coordinates can be reversed. EXHAUSTIVE: at each setting the sixteen systems RECOVER lists are exactly the sixteen this description generates, and the fibre identities and the binding pattern of the tree are as stated.

The cost. $\sum_i |A_i|! = 6+2+6+24+6+2+24+24 = 94$ against $\prod_i |A_i|! = 11,943,936$: by Theorem 5 the memoised RECOVER evaluates EXTEND at most 94 times on Λ , against the product for enumeration — five orders of magnitude, which is what tree structure buys. RECOVER tries at most $|A_r|! = 6$ root orders.

§6 · The reorderability law

6.1 The constraint system

Fix X with every order unknown, and let the orientation variables be as in D9. A pair of cells x, y differing on the k coordinates $I(x, y)$ constrains the orientation variables $o_i(x_i, y_i)$ for $i \in I$, and on nothing else: the meet and the join of x and y differ from x and y only on I .

Lemma 6 (complement closure). For every pair $x, y \in X$, the relation $C(x, y)$ of D9 is invariant under $\sigma \mapsto \sigma'$, the reversal of all its own orientations, and contains the all-ones and all-zeros assignments — it is a pair constraint in the sense of D9.

Proof. Reversing every orientation in I exchanges the meet of x and y with their join, and $C(x, y)$ requires both to be cells, a condition symmetric in the two. The all-ones assignment — x_i before y_i for every $i \in I$ — makes the meet x and the join y , both cells; the all-zeros assignment is its complement. ■ **PROVED**, and **EXHAUSTIVE**: over every subset of the boxes $2 \times 2 \times 2$ and $2 \times 2 \times 2 \times 2$ and every pair of cells in each — **1,792** and **1,966,080** pair constraints — every one is complement-closed and contains both constant assignments.

Lemma 7 (the difference relation). A relation on $\{0,1\}^k$ invariant under global complement is determined by the $k - 1$ exclusive-or differences $\sigma_1 \oplus \sigma_t$, $t = 2, \dots, k$. Writing

$$T(x,y) = \{ (\sigma_1 \oplus \sigma_2, \dots, \sigma_1 \oplus \sigma_k) : \sigma \in C(x,y) \} \subseteq \{0,1\}^{k-1},$$

the map $C \mapsto T$ is a bijection between complement-closed subsets of $\{0,1\}^k$ and subsets of $\{0,1\}^{k-1}$, and $T(x,y)$ always contains the origin.

Proof. The map $\sigma \mapsto (\sigma_1, \sigma_1 \oplus \sigma_2, \dots, \sigma_1 \oplus \sigma_k)$ is a bijection of $\{0,1\}^k$ under which global complement becomes “flip the first entry and leave the rest”. A subset invariant under that is a union of pairs $\{0,1\} \times \{\tau\}$, that is, a full preimage of a subset of $\{0,1\}^{k-1}$. The origin is the image of the all-zeros assignment, which Lemma 6 puts in C . ■ **PROVED.**

So a pair of cells at arity k imposes a relation on $k - 1$ Boolean variables, and that relation contains the origin. T is the auxiliary object; C is the constraint, and the two are not interchangeable for the question of §6.2. Since two cells differ on at most d coordinates, and two cells of the full Boolean box 2^d differ on exactly d :

The arity bound. $k \leq d$, and $k = d$ is attained.

6.2 What arises, and where bijunctivity stops

Theorem 7 (every pair constraint arises). At every arity $k \geq 2$, every pair constraint — every complement-closed $C \subseteq \{0,1\}^k$ containing 0^k and 1^k , equivalently by Lemma 7 every subset of $\{0,1\}^{k-1}$ containing the origin — is $C(x,y)$ for some index X of k coordinates and some pair x, y of its cells.

Proof. Let C be such a relation and let $X = C$, read as a set of cells of the box $\{0,1\}^k$ with every coordinate in its natural order. Both 0^k and 1^k are cells, so every value of every coordinate is used and X is an index. Take $x = 0^k$ and $y = 1^k$: they differ on every coordinate, so $I(x,y) = \{1, \dots, k\}$. For $\sigma \in \{0,1\}^k$, the join determined by σ takes $y_i = 1$ exactly where $\sigma_i = 1$ and is therefore the point σ itself, and the meet is its complement $\bar{\sigma}$. Both lie in X exactly when $\sigma \in C$, since C is complement-closed. Hence $C(x,y) = C$, and $T(x,y)$ is the difference relation of C . ■ **PROVED; EXHAUSTIVE** on two fronts: the construction was run for every pair constraint at $k \leq 4$ — $2 + 8 + 128 = 138$ relations — and reproduced each; and in the box $2 \times 2 \times 2$ the pairs of cells of all 256 subsets realise, at arity 1, 2 and 3, exactly **1**, **2** and **8** distinct constraints — which is 2 to the power $(2^{k-1} - 1)$, every pair constraint — while in the box $2 \times 2 \times 2 \times 2$ the same census over all 65,536 subsets realises **1**, **2**, **8** and **128** at arities 1 to 4. The census visited 1,792 and 1,966,080 pair constraints respectively.

Theorem 8 (the reorderability law). A pair constraint $C \subseteq \{0,1\}^k$ is bijunctive if and only if it is a partition constraint: there is a partition of $\{1, \dots, k\}$ such that C is exactly the set of σ constant on every block. The bijunctive pair constraints at arity k are therefore in bijection with the partitions of a k -element set and number the Bell number $B(k)$:

arity k	pair constraints	of these, bijunctive	$B(k)$	on the difference relation T
2	2	2	2	2 of 2
3	8	5	5	8 of 8
4	128	15	15	73 of 128
5	32,768	52	52	1,442 of 32,768

Every pair constraint at arity 2 is bijunctive; at arity 3 the three constraints “ $\sigma_i = \sigma_j$ or $\sigma_i = \sigma_1$ ”, one for each choice of the distinguished coordinate i , are not. Combined with the arity bound, **an index of two coordinates imposes only bijunctive constraints on its orientation variables, and the first non-bijunctive constraint appears at $d = 3$.**

Proof. ($\Leftarrow$) Let C be the constancy set of a partition and let $a, b, c \in C$. For i, j in one block, $a_i = a_j$, $b_i = b_j$ and $c_i = c_j$, so $\text{maj}(a, b, c)_i = \text{maj}(a_i, b_i, c_i) = \text{maj}(a_j, b_j, c_j) = \text{maj}(a, b, c)_j$. The majority is constant on every block, hence in C .

($\Rightarrow$) Let C be bijunctive. By Schaefer (1978), CITED, C is the solution set of a conjunction of clauses of at most two literals over $\sigma_1, \dots, \sigma_k$. Let Γ be the set of *all* clauses of at most two literals that every element of C satisfies; C is the solution set of some subset of Γ , so the solution set of Γ lies inside C , and C lies inside it by the definition of Γ . So C is exactly the solution set of Γ . Now a unit clause σ_i is false at 0^k and $\neg\sigma_i$ is false at 1^k ; a clause $\sigma_i \vee \sigma_j$ is false at 0^k and $\neg\sigma_i \vee \neg\sigma_j$ is false at 1^k ; both constant assignments lie in C , so none of these is in Γ , and every clause of Γ has the mixed form $\sigma_i \vee \neg\sigma_j$ with $i \neq j$ (a clause $\sigma_i \vee \neg\sigma_i$ is a tautology and constrains nothing). If $\sigma_i \vee \neg\sigma_j$ is satisfied by every $\sigma \in C$ then, C being complement-closed, it is satisfied by every $\bar{\sigma}$, which says that $\neg\sigma_i \vee \sigma_j$ is satisfied by every $\sigma \in C$. So the clauses of Γ come in complementary pairs, and each pair $(\sigma_i \vee \neg\sigma_j) \wedge (\neg\sigma_i \vee \sigma_j)$ is the equality $\sigma_i = \sigma_j$. The solution set of a conjunction of equalities is the set of σ constant on every class of the equivalence relation those equalities generate — a partition constraint.

The count. Distinct partitions give distinct constraints: if i and j share a block of one partition and not of the other, the indicator of the second partition's block containing i is constant on the second's blocks but not on the first's. So the bijunctive pair constraints at arity k are exactly the $B(k)$ partition constraints. ■ **PROVED** for every k ; **EXHAUSTIVE** for the table: every pair constraint at $k = 2, 3, 4, 5$ was enumerated, its bijunctivity decided by the two-clause closure — a relation is bijunctive exactly when it equals the solution set of the two-clauses it satisfies — and, for $k \leq 4$, by majority-closure as well, the two deciding alike on every relation; every bijunctive one was verified to be the constancy set of its own equality pattern; and the counts are 2, 5, 15, 52.

The difference relation is not the constraint. Counted on T the figures are 2, 8, 73 and 1,442 (last column, EXHAUSTIVE): every difference relation at arity ≤ 3 is majority-closed, and the two censuses agree only at arity 2. The direction that holds is C bijunctive $\Rightarrow T$ bijunctive, because T is the section of C at $\sigma_1 = 0$ and a section of a majority-closed relation is majority-closed; the census confirms it (no C is bijunctive with T not). The converse fails from arity 3 on, and its failure is the point: a difference variable $\sigma_1 \oplus \sigma_t$ is an exclusive-or of two orientation variables, exclusive-or is not bijunctive (D10), so a two-clause on the differences is not a two-clause on the orientations, and bijunctivity of T has no two-satisfiability consequence for the constraint system D9 defines.

REFUTATION, with the witness printed. The six cells

$$X = \{ 000, 111, 001, 110, 010, 101 \}$$

form an index in the box $2 \times 2 \times 2$ using every value of every coordinate. At the pair $x = 000$, $y = 111$ — arity 3 — the constraint is $C(x, y) = X$ itself, which is “ $\sigma_1 = \sigma_2$ or $\sigma_1 = \sigma_3$ ”; and

$$\text{maj}(001, 010, 111) = 011 \notin C(x, y),$$

so $C(x, y)$ is not bijunctive, while its difference relation $T = \{00, 01, 10\}$ is majority-closed. EXHAUSTIVE (the constraint is read off the witness, not asserted of it). The same happens at arity 4: the six cells $\{0000, 1111, 1100, 0011, 1010, 0101\}$ of $2 \times 2 \times 2 \times 2$ realise at $(0000, 1111)$ the difference relation $T = \{000, 011, 101\}$, whose majority 001 it excludes — there not even T is bijunctive — and translating T by 110 (exclusive-or, a symmetry of the difference encoding) gives $\{110, 101, 011\}$, exactly-two-of-three, whose coordinatewise complement is exactly-one-of-three.

6.3 The census

EXHAUSTIVE over ten boxes. For each box, every subset was tested for closure under every ordering system, by relabelling the subset under each of the $\prod |A_i|!$ permutation tuples and testing closure under the natural order.

box	subsets	ordering systems	reorderable	growth step	largest proper re-orderable
2×2	16	4	16	1	3
2×3	64	12	64	1	5
2×4	256	48	256	1	7
3×3	512	36	506	1	8
3×4	4,096	144	3,772	1	11
4×4	65,536	576	47,416	1	15
2×2×2	256	8	158	2	6
2×2×3	4,096	24	1,342	2	10
2×3×3	262,144	72	20,068	2	16
2×2×2×2	65,536	16	3,290	4	12

Table 2. Every subset of each box, the empty set and the singletons included — each is closed under every ordering, which is why the reorderable column reads 16 of 16 at 2×2; the step ranges over the reorderable subsets with at least two cells, as D11 says. The growth step is 2^{d-2} in every one of the ten boxes: 1 at $d = 2$, 2 at $d = 3$, 4 at $d = 4$, and in every box it is attained at the full box: the last column is the largest reorderable proper subset, and box minus that column equals the step in every row. At the six two-dimensional boxes the largest proper reorderable subset is one cell short of the box — the corner cell $(m-1, 0)$ of an $m \times n$ box is doubly irreducible, and Lemma 9 with $a = b$ removes it — whereas a product of three or more chains has no doubly irreducible cell, so at 2×2×3 and 2×3×3 no single cell can be removed and the largest proper reorderable subset is two short; at the Boolean boxes it is $3 \cdot 2^{d-2}$ — 3, 6 and 12 — which is 2^{d-2} short.

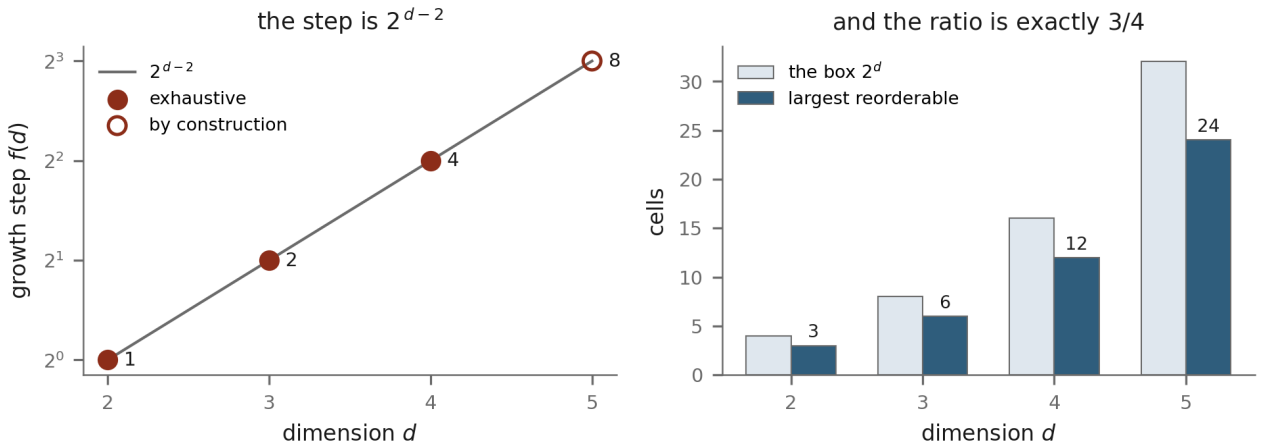


Figure 3. Left: the growth step against dimension. The filled markers at $d = 2, 3, 4$ are the exhaustive census of Table 2; the open marker at $d = 5$ is the drop at the full box, 8, which Theorem 9 supplies and the census does not reach — a lower bound on the step there, not the step. Right: the Boolean box against its largest reorderable proper subset — 3 of 4, 6 of 8, 12 of 16, 24 of 32 — exactly three quarters at every dimension.

6.4 The three-quarters theorem

Theorem 9 (proper sublattices of the Boolean box). For $d \geq 2$, every proper sublattice of $\{0,1\}^d$ has at most $3 \cdot 2^{d-2}$ cells, and that bound is attained. The cells removed by an attaining example form a subcube. Consequently drop at the full Boolean box is exactly 2^{d-2} , and $\text{step}(2^d) \geq 2^{d-2}$.

Proof. Attainment. Fix $i \neq j$ and let $S = \{ x : x_i \leq x_j \}$. If $x, y \in S$ then $(x \vee y)_i = \max(x_i, y_i) \leq \max(x_j, y_j) = (x \vee y)_j$ and likewise for the meet, so S is a sublattice; it excludes exactly the points with $x_i = 1, x_j = 0$, a subcube of size 2^{d-2} , so $|S| = 3 \cdot 2^{d-2}$ and S is proper.

The bound. Let $S \subsetneq \{0,1\}^d$ be a sublattice. If some coordinate takes only one value on S then S lies in a face and $|S| \leq 2^{d-1} = 2 \cdot 2^{d-2} \leq 3 \cdot 2^{d-2}$. Otherwise every alphabet of S is $\{0,1\}$, so D1's convention holds for S with the full alphabets. By Theorem 1 Step 4, S is the intersection of the cylinders over its pair projections. If every $\pi_{ij}(S)$ were the whole $\{0,1\}^2$, that intersection would be the whole box, contradicting properness. So some $\pi_{ij}(S)$ is a proper subset of $\{0,1\}^2$ and hence has at most 3 elements, whence $|S| \leq 3 \cdot 2^{d-2}$.

The drop. A relabelling of a two-value coordinate is either the identity or the transposition, so every relabelling of $\{0,1\}^d$ is a coordinate-flip pattern, and flips map the box to itself. Hence if $Y \subseteq \{0,1\}^d$ is reorderable — closed under the order some flip pattern induces — then the flipped image of Y is a sublattice of the same size, and conversely. So the largest proper *reorderable* subset of the box has the same size as the largest proper sublattice, $3 \cdot 2^{d-2}$; $\text{drop}(2^d) = 2^d - 3 \cdot 2^{d-2} = 2^{d-2}$, and $\text{step}(2^d) \geq \text{drop}(2^d)$ by D11. ■ **PROVED** for all $d \geq 2$.

MACHINE-CHECKED for $d = 2, \dots, 6$. Z3 was asked, over the 2 to the power 2^d candidate subsets, whether a proper sublattice with more than $3 \cdot 2^{d-2}$ cells exists (unsat at every d) and whether one with exactly $3 \cdot 2^{d-2}$ exists (sat at every d). The five obligations give the bounds 3, 6, 12, 24 and 48; the drop 1, 2, 4, 8, 16 is arithmetic on those answers and is recorded on the same five lines, not as an obligation of its own.

Corollary 2 (the maximal sublattices of the Boolean box). For $d \geq 2$, every maximal proper sublattice of $\{0,1\}^d$ is $\{ x : x_i \leq x_j \}$ for some $i \neq j$; each has exactly $3 \cdot 2^{d-2}$ cells, and there are $d(d-1)$ of them.

Proof. Each $S = \{ x : x_i \leq x_j \}$ is a proper sublattice of the largest size a proper sublattice can have (Theorem 9), so any sublattice strictly containing it is the whole box: S is maximal. Distinct pairs (i, j) remove distinct subcubes, so these are $d(d-1)$ distinct maximal sublattices. Conversely let K be a maximal proper sublattice. By Rival (1973), CITED — for a maximal proper sublattice K of a finite distributive lattice L , the complement $L \setminus K$ is an interval $[a, b]$ with a join-irreducible and b meet-irreducible, in the standard sense that excludes the bottom and the top — the removed set is $[a, b]$. In $\{0,1\}^d$ the join-irreducibles in that sense are the unit vectors e_i , since a point with two coordinates equal to 1 is the join of two smaller points, and dually the meet-irreducibles are the co-unit vectors $1 - e_j$. So $a = e_i$ and $b = 1 - e_j$, and $a \leq b$ forces $i \neq j$; then $[a, b] = \{ x : x_i = 1, x_j = 0 \}$ and $K = \{ x : x_i \leq x_j \}$. ■ **PROVED; EXHAUSTIVE** for $d = 2, 3, 4$ over every sublattice of the box: 2, 6 and 12 maximal proper sublattices, every one of 3, 6 and 12 cells.

Rival's theorem is a statement about *maximal* sublattices, and the cardinality bound that accompanies it, $|L| \leq (3/2)|K|$, bounds a maximal sublattice from *below* — tight on the three-element chain, whose three maximal proper sublattices have two of its three elements (EXHAUSTIVE). Theorem 9 bounds every proper sublattice of the Boolean box from *above*. The two are not comparable; what Corollary 2 says is that in the Boolean box every maximal sublattice is exactly three quarters, strictly inside the two thirds that Rival's bound allows in general.

6.5 What the law does not establish

Theorem 8 places the pair-constraint language outside the bijunctive class at arity 3. Schaefer's dichotomy (1978), CITED, is the reason that matters, and it is also the reason for care: over the Boolean domain a constraint language is tractable when every relation is 0-valid, or every one is 1-valid, or Horn, or dual Horn, or bijunctive, or affine, and satisfiability is NP-complete for every other language.

The pair constraints alone are trivially satisfiable. Every $C(x,y)$ contains 0^k and 1^k (Lemma 6), so the language of pair constraints on its own is 0-valid and 1-valid, two of the six classes: the all-zeros assignment satisfies every pair constraint of every index at once. What is neither 0-valid nor 1-valid is antisymmetry, $o_i(u,v) \neq o_i(v,u)$, and it is antisymmetry that makes the pair constraints a problem. The arity-3 witness $C_3 = \sigma_1 = \sigma_2$ or $\sigma_1 = \sigma_3$ of §6.2 is

closed under neither majority, nor conjunction, nor disjunction, nor ternary exclusive-or (EXHAUSTIVE), so it is neither bijunctive, Horn, dual Horn nor affine; the language $\{C_3, \neq\}$ lies in none of the six classes, and satisfiability for it is NP-complete. That is a statement about a language, obtained at arity 3 because the constraint is C and not T .

This does not decide the complexity of reorderability. Three things stand between the two statements, and this paper resolves none of them.

1. **Realisability.** To conclude hardness one must map an arbitrary instance of the hard language to an index whose pairs impose exactly the wanted constraints. The map from an index to its constraint system is global — adding one cell changes the constraints of every pair it forms — non-separable and non-monotone, and no reduction is given here.
2. **The order axioms.** The orientation variables must also satisfy transitivity on each coordinate, a three-literal clause. Bijunctivity of the pair constraints therefore does not by itself make the decision a two-satisfiability instance even at $d = 2$, where every pair constraint is bijunctive. What decides the two-coordinate case is Theorem 2, at the cost stated in §3.3.
3. **Sparsity is not available in general.** The variables are the $\sum_i |A_i|(|A_i| - 1)$ orientation variables and the constraints are indexed by pairs of cells, with scope the coordinates $I(x,y)$ on which the pair differs; every subset of coordinates up to the full set can occur as a scope, so the constraint hypergraph on the d coordinates is complete, its treewidth is $d - 1$, and a tree decomposition along it returns only the enumeration bound $\prod_i |A_i|!$. The structure Theorem 5 exploits is different — it is a hypothesis on X , that X is the intersection of cylinders over the edges of a tree — and it is not a property of the constraint hypergraph.

Where the problem sits in the literature. Jeavons and Cooper (1995), CITED, prove that *max-closed* constraints — relations closed under the coordinatewise maximum in a given order on each domain — form a tractable class. Green and Cohen (2008), CITED, ask exactly the question of this section for that class: whether there exist permutations of the domains, one per variable, under which every constraint of an instance becomes max-closed — the *domain permutation reduction* — and show that finding them is tractable over a Boolean domain and NP-hard in general. Reorderability asks for closure under both minimum and maximum of a single relation, the index, with an order per coordinate; in the T-structured case it is a binary instance whose constraints are the edge projections, each on its own pair of domains. Neither result settles it: the hardness of Green and Cohen is for max-closure alone, of arbitrary sets of constraints over a common domain, and their Boolean tractability is for a common Boolean domain, where a reordering is a pattern of coordinate flips and the question is Theorem 9’s setting. “Open” here means open relative to those two results.

The honest summary is: **the pair-constraint language leaves the bijunctive class at arity 3, which is at $d = 3$, and with antisymmetry it leaves every tractable class there; the complexity of reorderability itself is open here.**

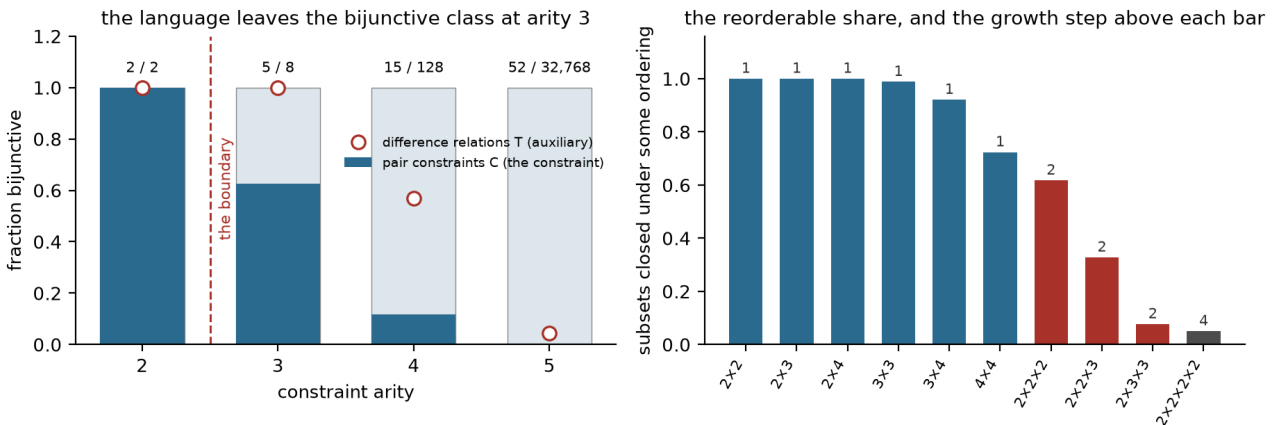


Figure 4. Left: for each arity, the fraction of the pair constraints — complement-closed relations on the orientation variables containing both constant assignments — that are bijunctive: 2 of 2 at arity 2, 5 of 8 at arity 3, 15 of 128 at arity 4, 52 of 32,768 at arity 5, the Bell numbers of Theorem 8. The hollow markers are the same fraction counted on the difference relation T instead — 2 of 2, 8 of 8, 73 of 128, 1,442 of 32,768 — which is not the constraint. The boundary lies between arity 2 and arity 3. Right: for each of the ten boxes of Table 2, the fraction of subsets that are reorderable, with the growth step printed above the bar. The fraction falls with the number of coordinates: 1.000 at 2×2 , 0.724 at 4×4 , 0.077 at $2 \times 3 \times 3$, 0.050 at $2 \times 2 \times 2 \times 2$.

§7 · What an index defends

A closed index expresses itself (Corollary 1). This section states what that expression is good for and, as precisely, what it is not.

7.1 Membership is decided, not estimated

Corollary 1 makes the membership predicate total on the ambient box. That is a stronger property than accuracy: a total predicate has no value outside $\{0,1\}$, so it cannot return “undetermined” and have a caller coerce that to “false”. On Λ , both the conjunction of the seven bounds and the conjunction of the 56 envelope comparisons were evaluated at **all 6,912** ambient points and returned a value at every one, agreeing with the cell list at every one — EXHAUSTIVE.

7.2 Derived quantities, and the counting bound

Let an index assign parameters $p \in \mathbb{R}^a$ to each cell and let $q = \Phi(p) \in \mathbb{R}^b$ be quantities derived from them by a differentiable map. Write $D = b - \text{rank}(\partial\Phi/\partial p)$, the rank taken at a point; where the rank is locally constant, D is the number of functionally independent relations the derived quantities must satisfy near that point.

Lemma 8. $D \geq b - a$. In particular if $b > a$ then $D \geq 1$.

Proof. $\partial\Phi/\partial p$ is a $b \times a$ matrix; the rank of a matrix exceeds neither of its dimensions, so $\text{rank} \leq a$ and $D = b - \text{rank} \geq b - a$. ■ **PROVED**, with no closure and no smoothness beyond differentiability. **SAMPLED:** 60 pseudorandom quadratic maps with integer coefficients, seed 3, Jacobians evaluated at rational points and their ranks computed in exact rational arithmetic by fraction-free elimination; $\text{rank} \leq \min(a,b)$ and $D \geq b - a$ in all 60. The sample corroborates the implementation, not the lemma, which is proved.

The bound is old. Edlén (1964, §17, footnote), discussing term-value formulae, declines a least-squares fit of a series formula’s parameters to the levels of that series on the ground that with as many parameters as observations “no check is possible” — which is $D = 0$ in the case $b = a$. The remark is quoted here at second hand. The general form, and the observation that the *rank* and not the dimension is what binds, is what Lemma 8 adds.

7.3 What a fabricated cell costs

Let y be an ambient non-cell of a closed X . The closure of $X \cup \{y\}$ must admit y and everything the envelopes then force; write

$$A(y) = |\mathcal{R}(X \cup \{y\})| - |X| - 1$$

for the number of *further* cells forced, beyond y itself. On Λ an ambient non-cell is a label the seven bounds refuse — for instance any label with a 1p source subshell, $n = 1$ and $\ell = 1$, which violates $\ell \leq n - 1$.

EXHAUSTIVE on Λ . Over **all 5,936** ambient non-cells at the larger cap setting: minimum **15**, median **309**, mean **380.9**, maximum **1,795**, and **not one** value is zero. The minimum is attained at 32 non-cells, the first of them (2, 1, 3, 2, 2, 1, 3, 0) — a label in which three electrons arrive at the target where two were removed, $g = 3 > q = 2$. There is no ambient point of Λ 's box that can be inserted without the envelopes forcing further cells.

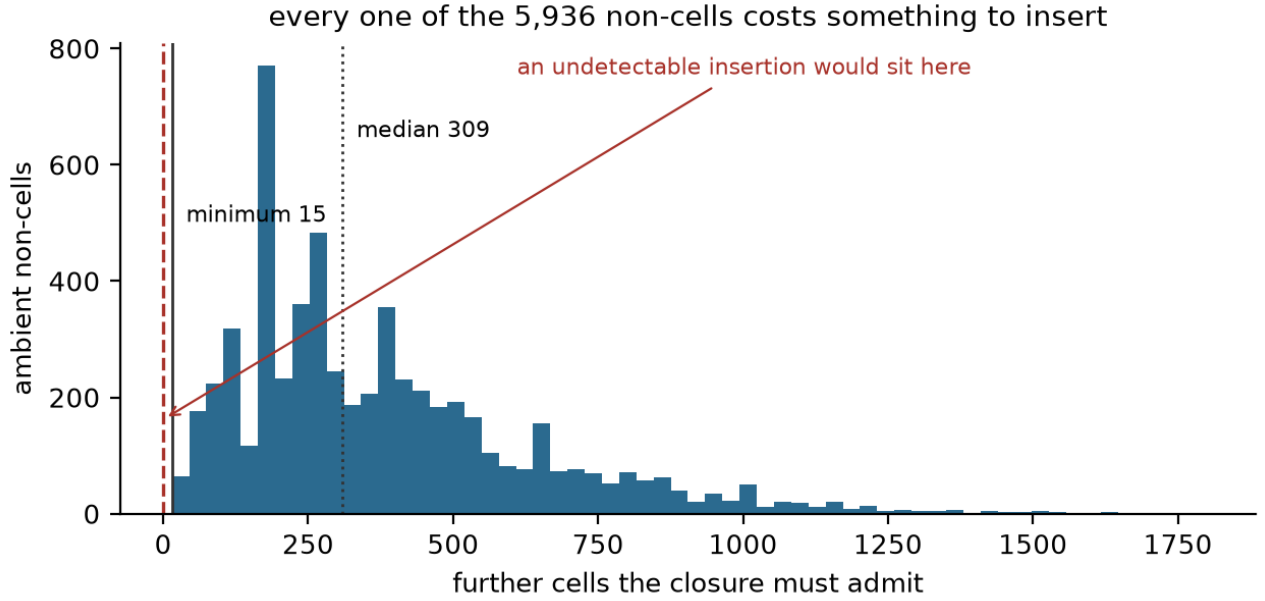


Figure 5. The cost of inserting one ambient non-cell into Λ , over every one of the 5,936 of them. The dashed line at zero marks where a free insertion would sit; no insertion reaches it. The minimum is 15 and the median 309.

What this does not defend against. $A(y)$ prices *insertion into* an index. It does not price *replacement of* one. An index rebuilt around y from the start, rather than asked about y , is closed, has $E = 0$ and satisfies every result of this paper; it is a different index and there is nothing in the structure that distinguishes it from the original. The defence is available only to someone who asks the index before extending it. Nor does it price a label that is admissible but unrealised — the wrong-parity cells of §5.1 are cells, and A does not see them.

7.4 What cannot be removed

Lemma 9 (interval removal; the maximal case is Rival 1973, CITED). Let X be closed and $a \leq b$ cells of X . Then $X \setminus [a, b]$ is closed if and only if a is join-prime in X and b is meet-prime.

Here join-prime includes the bottom cell and meet-prime the top (§1, Notation): with a the bottom, $a \leq x$ holds for every x , the join-prime condition is vacuous, and the lemma says that a down-set $[bottom, b]$ can be removed exactly when b is meet-prime. In a distributive lattice join-irreducible and join-prime coincide (Birkhoff 1967, CITED), so the hypothesis is a condition on irreducibles. For a *maximal* proper sublattice K of a finite distributive lattice, Rival (1973) proves that the complement is such an interval, with a join-irreducible and b meet-irreducible; what the lemma adds is the biconditional for every interval of a closed index, not only for the complement of a maximal sublattice.

Proof. Write $R = [a, b] = \{ c \in X : a \leq c \leq b \}$. Suppose a is join-prime and b meet-prime, and let $x, y \in X \setminus R$ with $x \vee y \in R$. Then $a \leq x \vee y$, so by join-primality $a \leq x$ or $a \leq y$; say $a \leq x$. Also $x \leq x \vee y \leq b$. So $a \leq x \leq b$ and $x \in R$, a contradiction. Dually with meet-primality and b . So the removal is closed.

Conversely suppose a is not join-prime. By the coincidence of prime and irreducible elements in a distributive lattice (§1, Notation), a is not join-irreducible: $a = u \vee v$ with $u, v \in X$, neither above a . Both are $\leq a \leq b$ and neither is in R (being in R requires being $\geq a$), while $u \vee v = a \in R$. So the removal is not closed. Dually for b . ■ **PROVED.**

MACHINE-CHECKED. Z3 verified the biconditional over every sublattice S of three boxes and every pair $a \leq b$ in each: 3×3 (36 pairs, 2^9 candidate subsets per pair), $2 \times 2 \times 2$ (27 pairs, 2^8), $2 \times 2 \times 3$ (54 pairs, 2^{12}). No failure.

On Λ . The 18 join-prime cells and the 18 meet-prime cells have **empty intersection**, so no interval $[a, a]$ — no single cell — can be removed, at either cap setting. The smallest removable interval has **4** cells, running from $(2, 1, 3, 3, 2, 1, 3, 0)$ to $(3, 1, 3, 3, 3, 1, 3, 0)$, a box free in exactly the two shell coordinates; removing it leaves 972 cells with zero meet or join failures across all **471,906** surviving pairs. The same fact counted the other way: for each cell, count the pairs of *other* cells whose meet or join is that cell. Over all **475,800** pairs of Λ the minimum over cells is **16**, the median 503.5, the maximum 5,091, the mean 739.0, and **no cell** has count zero — a cell with count zero would have to be join-irreducible and meet-irreducible at once, and there is none.

7.5 The three properties, and their scope

property	what it gives	what it cannot give
alphabet (projection)	the values, always	nothing about their order
bounds (envelopes)	the index, from Corollary 1, whenever it is closed	a bound of arity above 2
order (Theorems 2, 5)	every admissible ordering system, at $d = 2$ always and above 2 when the constraint graph is a tree	uniqueness — only up to the ties of $\sqsubseteq$, and up to reversal of each component of the tree

§8 · Verification record

A single verification program recomputes every finite claim of this paper. It uses the Python standard library, Z3 (de Moura and Bjørner 2008, CITED) and NumPy, imports its reference closure operator and its index fixtures from the implementations that define them rather than copying them, and prints one line per obligation. On the run reported here: Z3 5.1.0, NumPy 2.5.2, Python 3.12.3.

The two guards, run before any obligation is reported. No Z3 result below is printed unless every line of both passes. The non-vacuity guard runs on every box a Z3 obligation ranges over; the fidelity guard evaluates every Z3 encoding the obligations use — none is left unguarded — at concrete instances against a reference written from the paper’s definitions.

guard	what it checks	result
non-vacuity, Theorems 2 and 3	“X uses every value, s a total order, X closed under s , X a proper subset” is satisfiable	satisfiable in 3×3 , 4×4 , $2 \times 2 \times 3$, $3 \times 3 \times 3$, 3×5 , 5×4
non-vacuity, Lemma 9	“S closed, a and b in S, S a proper subset” is satisfiable	satisfiable in 3×3 , $2 \times 2 \times 2$, $2 \times 2 \times 3$
non-vacuity, Theorem 9	“S closed, non-empty, a proper subset” is satisfiable	satisfiable in 2^d , $d = 2, \dots, 6$
fidelity (a)	the encoding of closure-under- s equals an independent closure test	150 instances, 0 disagreements
fidelity (b)	the encoding of $\sqsubseteq$ equals the concrete relation	1,387 ordered pairs, 0 disagreements
fidelity (c)	the encoding of the recovered order s^* equals the sort of D_7	944 ordered pairs, 0 disagreements
fidelity (d)	the harness’s closure predicate (Theorem 9, Lemma 9) equals a concrete closure test	150 instances, 0 disagreements

guard	what it checks	result
fidelity (e)	Theorem 3's encodings — fibres are intervals; lo and hi comparisons; endpoints monotone along s — equal the concrete tests	120 instances; 3,700 endpoint comparisons; 120 instances; 0 disagreements
fidelity (f)	Lemma 9's encodings — S minus an interval closed; a join-prime; b meet-prime — equal the concrete tests	120 instances, 0 disagreements
reference closure (g)	“closed under the natural order” equals “the reference closure operator adds nothing”	120 instances, 0 disagreements
reference closure (h)	the fast local envelope routine equals the reference operator, cell for cell	120 instances, 0 disagreements
negative control on the guard	a deliberately wrong reference is detected by the fidelity comparison	detected, 10 disagreements

Obligations by status.

§	object	status	family or box
2.1	Theorem 1, the closure equivalence	PROVED	—
2.1	Theorem 1 and $\mathcal{R}(X)$ = the pair-projection intersection	EXHAUSTIVE	every subset of 3×3 , 2×4 , $2 \times 2 \times 2$, $2 \times 2 \times 3$, $2 \times 2 \times 2 \times 2$ — 70,598
2.1	Corollary 1, totality of membership: the seven bounds, and the envelope conjunction	PROVED; EXHAUSTIVE	all 6,912 ambient points of Λ , both predicates
2.2	Lemma 1, necessity of “may precede”	PROVED	—
2.2	Lemma 2, reflexivity and totality on a closed index	PROVED	—
2.2	Lemma 3, transitivity of $\sqsubseteq$ for every index	PROVED	—
2.3	Theorem 2 (a), (b) both directions, (c)	MACHINE-CHECKED	3×3 ($2^9 \times 6$), 4×4 ($2^{16} \times 24$), $2 \times 2 \times 3$ ($2^{12} \times 2$), $3 \times 3 \times 3$ ($2^{27} \times 6$) — 16 obligations
2.3	$\sqsubseteq$ need not be antisymmetric	REFUTATION	witness in each of the four boxes
2.3	Lemma 4, the extensions are the answer	PROVED	—
3.1	Theorem 3, the fibre characterisation	PROVED; MACHINE-CHECKED	3×3 , 4×4 , 3×5 , 5×4 — 4 obligations
3.1	Theorem 3 for a declared alphabet	REFUTATION	the four-cell witness with A_1 declared $\{0,1,2\}$
3.2	Lemma 5, the nesting lemma	PROVED; EXHAUSTIVE	2,354 families of distinct intervals; 5,084 multisets
3.2	“no nesting at all” as the criterion	REFUTATION	fails on 1,098 of the 2,354
3.3	DECIDE ₂ against brute force	EXHAUSTIVE	44,033 subsets over five boxes
3.3	the worked example of Figure 1	EXHAUSTIVE	all 2,880 ordering systems of the 5×4 box

§	object	status	family or box
4.1	every pair projection reorderable does not make X reorderable	PROVED (the three-cell witness); EXHAUSTIVE	193, 3,271 and 63,775 subsets of $2 \times 2 \times 2$, $2 \times 2 \times 3$, $2 \times 2 \times 2 \times 2$ using every value
4.2	Theorem 4, factorisation over a graph	PROVED	—
4.3	Theorem 5, RECOVER returns every solution, at the stated cost	PROVED; EXHAUSTIVE	7,486 tree-structured subsets; 362,144 scrambles
4.4	Theorem 6, the diagnostic split	PROVED; EXHAUSTIVE	the same 70,598 subsets; 0 with neither witness
4.4	periodic layout $E = 36$; left-step $E = 0$ on the occupied subshells	EXHAUSTIVE	90 cells in 126; 19 and 20 subshells in 32
5.1	Λ : 976 cells, box 6,912, $E = 0$; 216 at the smaller caps; 413 wrong-parity cells	EXHAUSTIVE	both cap settings
5.2	Λ : 18 join- and 18 meet-primes, intersection empty; 16 binding pairs; tree-structured	EXHAUSTIVE	all 6,912 ambient points
5.3	Λ recovered from a scrambled bag	SAMPLED	20 scrambles per cap setting, seed 2026
5.3	Λ admits exactly 16 ordering systems, and which sixteen	EXHAUSTIVE	every solution of RECOVER, both settings, against the described sets
6.1	Lemma 6, complement closure	PROVED; EXHAUSTIVE	1,792 and 1,966,080 pair constraints
6.1	Lemma 7, the difference relation	PROVED	—
6.2	Theorem 7, every pair constraint arises	PROVED; EXHAUSTIVE	the construction at every constraint of arity ≤ 4 (138); every subset of $2 \times 2 \times 2$ and of $2 \times 2 \times 2 \times 2$
6.2	Theorem 8, bijunctive = partition constraint, Bell numbers	PROVED; EXHAUSTIVE	every pair constraint at arity 2, 3, 4, 5 — 2, 8, 128, 32,768; majority and two-clause closure agree at arity ≤ 4
6.2	the census on the difference relation T, and C bijunctive $\Rightarrow$ T bijunctive	EXHAUSTIVE	every relation on 1, 2, 3, 4 difference variables
6.2	a non-bijunctive pair constraint at arity 3	REFUTATION	the six-cell witness in $2 \times 2 \times 2$, constraint read off it
6.2	a non-bijunctive difference relation at arity 4	REFUTATION	the six-cell witness in $2 \times 2 \times 2 \times 2$, relation read off it
6.3	the census, the growth step, and its attainment at the full box	EXHAUSTIVE	ten boxes, every subset of each
6.4	Theorem 9, three quarters	PROVED; MACHINE-CHECKED	$d = 2, \dots, 6$, 2 to the power 2^d candidates each — 5 obligations
6.4	Corollary 2, every maximal sublattice	PROVED (with Rival 1973, CITED); EXHAUSTIVE	every sublattice of $2^2, 2^3, 2^4$; the three-element chain
6.5	the arity-3 witness lies in no Schaefer class with $\neq$	EXHAUSTIVE	the four closure tests on the witness
7.2	Lemma 8, the counting bound	PROVED; SAMPLED	60 maps, seed 3, exact rational rank

§	object	status	family or box
7.3	$A(y) > 0$ for every ambient non-cell of Λ	EXHAUSTIVE	all 5,936
7.4	Lemma 9, interval removal	PROVED; MACHINE-CHECKED	3×3 , $2 \times 2 \times 2$, $2 \times 2 \times 3$, every pair $a \leq b$ — 3 obligations
7.4	Λ 's removal minimum 4; push-back minimum 16	EXHAUSTIVE	475,800 pairs, 471,906 surviving pairs

Totals. 111 obligations: 72 EXHAUSTIVE, 28 MACHINE-CHECKED, 8 REFUTATION, 3 SAMPLED, beside the proofs written out above. Every number printed in this paper is asserted by the line that prints it, so that a drift in any of them fails the run. Five negative controls run beside the obligations, each of which must be reported refuted for the run to count as evidence: the false antisymmetry claim, the weak reading of the nesting lemma, the false growth step 2^{d-1} , “every arity-4 difference relation is bijunctive”, and “every arity-3 pair constraint is bijunctive”.

What is not machine-checked, and why. Theorems 1, 4, 5, 7, 8, 9 and Lemmas 1–9 are proved for general d and for alphabets of any size; a Z3 obligation ranges over a fixed finite box and cannot express them. What Z3 contributes is a check of the proofs’ instances over every subset of a box that uses every value and, where an order is unknown, every total order on the named coordinate — a quantification a sampled sweep cannot supply. The growth step at $d \geq 5$ over all reorderable sets, and the complexity of reorderability, are the two open items and are stated as such in §0 and §6.5.

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